Enhanced dissipation for passive scalars

12 November 2026

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1 Overview

Discuss on Zulip: #overview, topic “Overview”.

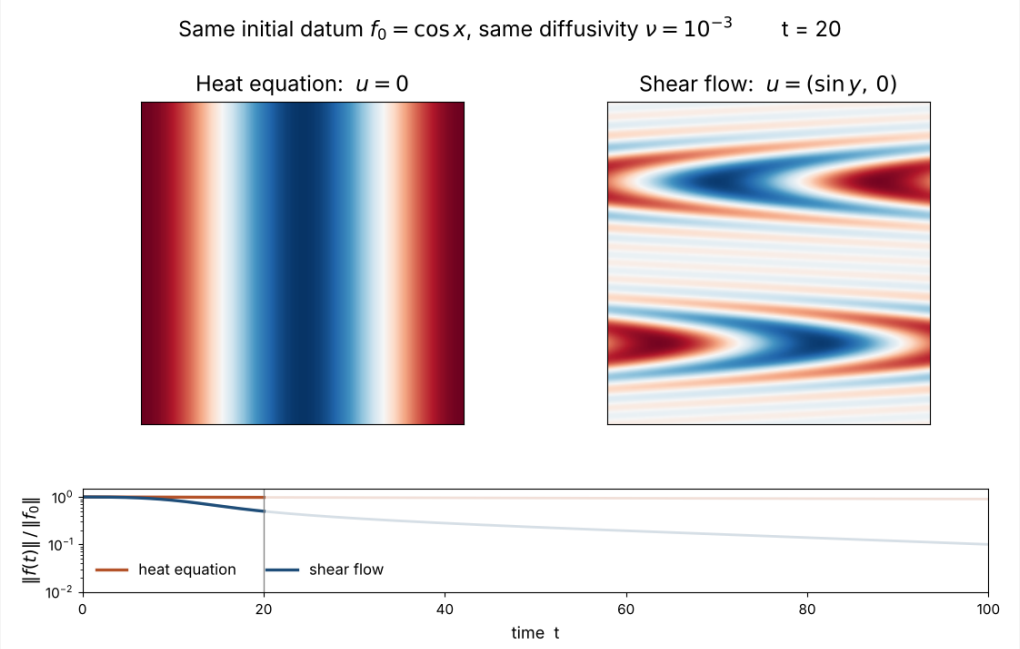
A passive scalar is a quantity carried by a fluid without acting on it, such as temperature, a dye in water or a pollutant in the air. If f is its concentration, u the velocity of an incompressible flow, and $\nu > 0$ the molecular diffusivity, then f solves the advection–diffusion equation

$$\partial_t f + u \cdot \nabla f = \nu \Delta f, \qquad \nabla \cdot u = 0.$$

Diffusion alone dissipates the scalar on a time scale of order $1/\nu$, which is very long when ν is small. The flow, on the other hand, dissipates nothing: it only rearranges the scalar. In doing so, it can transfer large-scale structures to small scales, for instance by stretching the scalar into thin filaments, and at small scales diffusion is much more efficient. The result is that f decays on a time scale much shorter than $1/\nu$: this is *enhanced dissipation*.

How, and by how much, does an incompressible velocity field speed up the dissipation of a diffusing scalar?

The two videos below show the phenomenon in the simplest setting, a periodic box in two dimensions. In the first, a steady shear flow significantly accelerates the dissipation of vertical stripes, orthogonal to its streamlines.

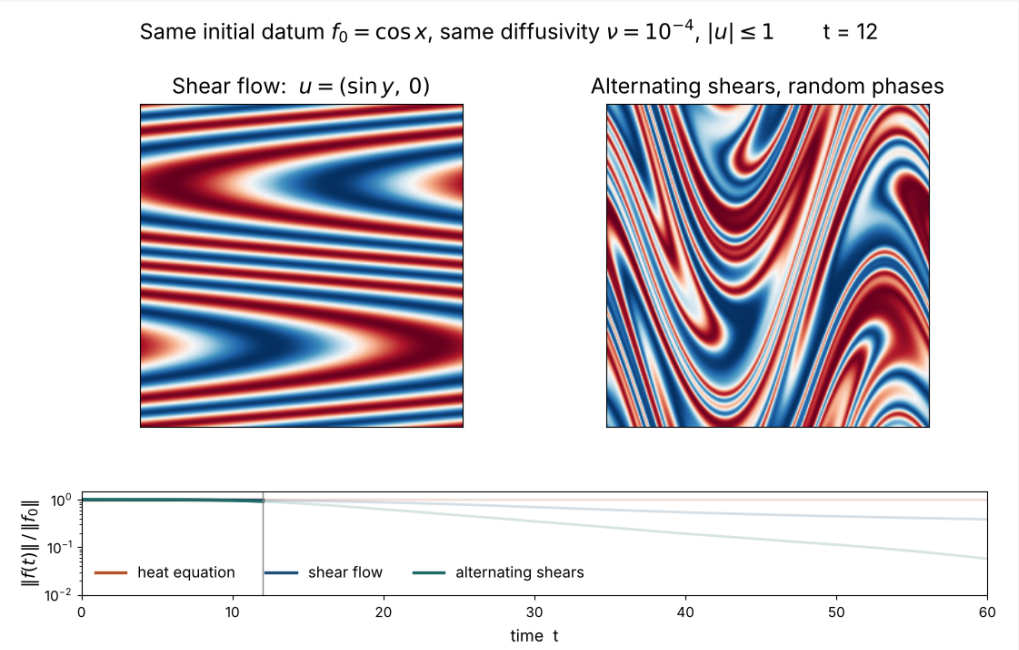


Video. Left: the heat equation. Right: the same initial datum carried by the shear flow $u = (\sin y, 0)$, with the same diffusivity. Bottom: the size of the two solutions in time, on a logarithmic scale. [numerics]
The video plays in the online edition of this page. How it was made: page Numerical methods, record `num-heat-vs-shear`.

The second video compares two flows of the same size. A steady shear stretches the scalar only along its own streamlines. Shears that alternate in direction, with random phases (the model of Pierrehumbert), stretch and fold the scalar, which then dissipates much faster and more uniformly in space.

Quantifying the time scale of enhanced dissipation is the subject of this wiki. Laminar flows, such as shear flows, typically dissipate the scalar on time scales of order ν^{-a} with $0 < a < 1$, whereas *mixing* flows, such as the alternating shears, can dissipate it on time scales of order $|\log \nu|$. The questions are which time scales are possible, which flows achieve them, and how one proves it.

Mixing: the problem without diffusion. When $\nu = 0$ the equation is pure transport: the flow conserves every L^p norm of f and dissipates nothing. It can still *mix*, that is, move the scalar to ever smaller scales, so that f converges weakly to its mean while its L^2 norm stays constant. The speed of mixing can be measured by the decay of a negative Sobolev



Video. Left: the shear flow $u = (\sin y, 0)$. Right: shears that alternate, horizontal and vertical, each acting for 3 units of time, with random phases. Same initial datum $f_0 = \cos x$, same diffusivity $\nu = 10^{-4}$; both velocity fields have size at most 1. Bottom: the size of the solutions in time, on a logarithmic scale. [numerics]
The video plays in the online edition of this page. How it was made: page Numerical methods, record `num-alternating-shears`.

norm, such as $\|f\|_{H^{-1}}$. Mixing is a field of its own, with its own questions and methods; this wiki covers only its relation to enhanced dissipation.

Beyond passive scalars. The same mechanism, acting on the vorticity of a two-dimensional fluid, helps keep shear flows stable at large Reynolds numbers, against perturbations that are small enough in terms of the Reynolds number; the page Why it matters expands on this.

The setting, made precise. In most of the wiki, the domains without boundary are the torus $\mathbb{T}^2$ and the periodic strip $\mathbb{T} \times \mathbb{R}$; some problems are set in the whole plane $\mathbb{R}^2$. The domains with boundary are the disk and the periodic channel $\mathbb{T} \times [0, 1]$; the boundary conditions are stated where they are used. The initial datum f_0 is in L^2 and has mean zero; for shear and circular flows, it also has zero average along each streamline, since the flow does not act on that average. The size of a solution is its L^2 norm. One definition of the dissipation time of a flow is the time it needs to reduce every initial datum by a fixed factor, say one half; the flow enhances dissipation when this time is much shorter than $1/\nu$. Definitions and notation are collected in Notation.

1.1 Map of the wiki

- *Motivation and history*: why the problem matters and where it comes from.
- *Techniques through examples*: enhanced dissipation for two flows, Couette and Poiseuille,

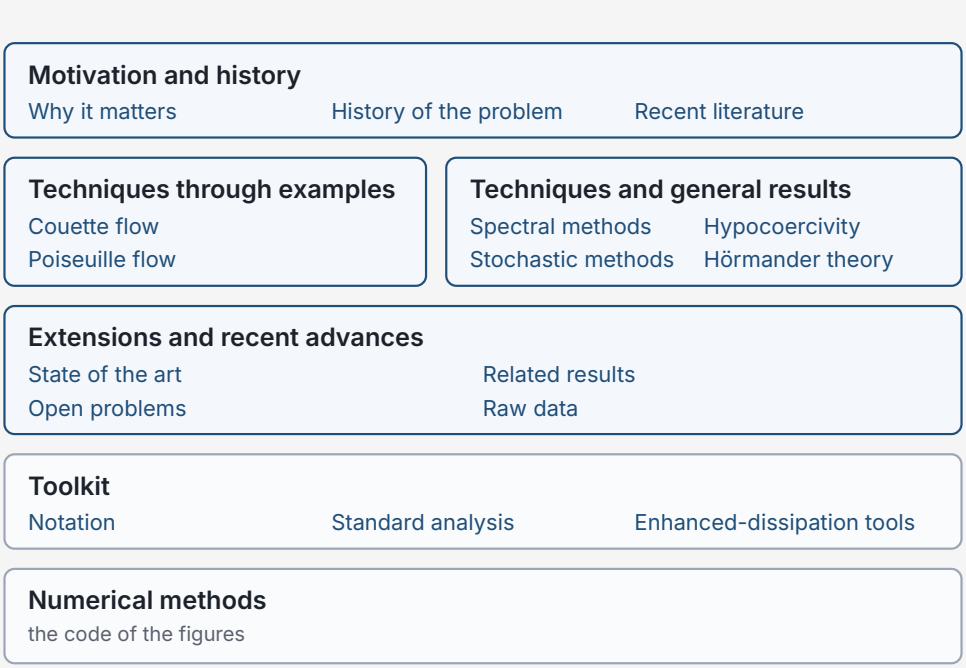


Diagram. The parts of the wiki; click a name to open its page.

proved once with each method, so that the methods can be compared on the same example.

- *Techniques and general results*: the general theorem behind each method, with its proof, for spectral methods, hypocoercivity, stochastic methods and Hörmander theory.
- *Extensions and recent advances*: the state of the art, related results and open problems.
- *Toolkit*: the notation and the standard analysis that the proofs use.
- *Numerical methods*: the codes behind the pictures.
- *About this wiki*: how to contribute and who did what.

Part I
Motivation and history

2 Why it matters

Contributors: to be named · Status: skeleton
Discuss on Zulip: #overview, topic “Why it matters”.

[Why the problem matters, in mathematics and in applications, written as the introduction of a paper.]

3 History of the problem

Contributors: to be named · Status: skeleton
Discuss on Zulip: #overview, topic “History of the problem”.

[A short account of the progress; then the history technique by technique.]

3.1 Origins

[Before the mathematical theory.]

3.2 Spectral methods

[Contributions and dates; the general theory is on Spectral methods.]

3.3 Hypocoercivity

[Contributions and dates; the general theory is on Hypocoercivity.]

3.4 Stochastic methods

[Contributions and dates; the general theory is on Stochastic methods.]

3.5 Hörmander theory

[Contributions and dates; the general theory is on Hörmander theory.]

4 Recent literature

Contributors: to be named · Status: skeleton
Discuss on Zulip: #overview, topic "Recent literature".

[The latest papers, with what each contributes; their results are collected on State of the art and Related results.]

Part II
Techniques through examples

5 Couette flow

Contributors: Michele Dolce · Status: skeleton
Discuss on Zulip: #examples, topic "Couette flow".

The Couette flow is the linear shear $u = (y, 0)$, considered here in the simplest possible domain for this problem, $(x, y) \in \mathbb{T} \times \mathbb{R}$. The equation under study then reduces to

$$\partial_t f + y\partial_x f = \nu \Delta f, \qquad f|_{t=0} = f^{\mathrm{in}}, \qquad (x, y) \in \mathbb{T} \times \mathbb{R}.$$

This is a foundational model in the field, because the same equation governs the vorticity of the two-dimensional Navier–Stokes equations linearized around the Couette flow. In that context Kelvin already solved it explicitly in 1887 (see Kelvin (1887, eq. (34)–(35))). Moreover, the same model, without the diffusion in x, was derived by Kolmogorov (1934) when developing his theory of Brownian motion, and the equation is also of Fokker–Planck type. See also History of the problem.

We use this model to illustrate what typical mixing and enhanced dissipation results look like, and how the two are related. The model is simple enough that each proof technique can be carried out in full. They are all presented below on this single example.

Why this domain? The domain $\mathbb{T} \times \mathbb{R}$ is mathematically convenient: there are no boundaries, and one can take the Fourier transform in both variables.

There is also a more geometric motivation. Consider a circular flow in the plane, $u = rV(r)e_\theta$, for which the equation reads $\partial_t f + V(r)\partial_\theta f = \nu\Delta f$. The log-polar coordinates $X = e^s\cos(\theta)$, $Y = e^s\sin(\theta)$ identify $\mathbb{R}^2 \setminus \{0\}$ with $(\theta, s) \in \mathbb{T}\times\mathbb{R}$, as shown in the figure below.

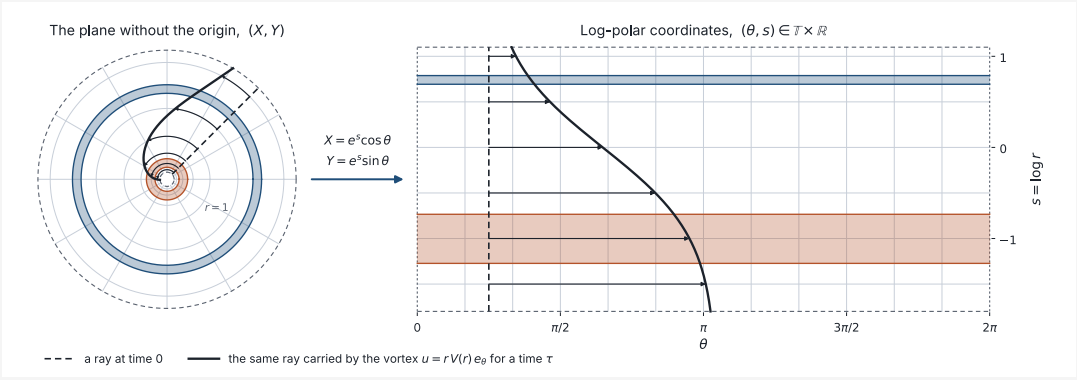
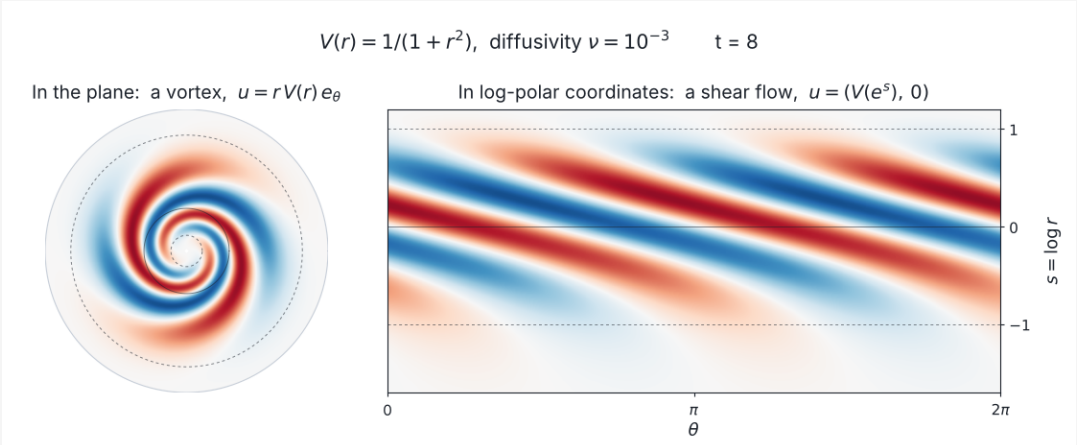


Figure. Left: the plane, with two annuli of the same width and a ray (dashed) carried by a circular flow for some time (solid). Right: the same picture in log-polar coordinates. The annuli become strips, and the circular flow becomes a shear flow.

Note that this is a conformal change of variables, so the Laplacian only gains a scalar factor, and the equation becomes

$$\partial_t f + V(e^s)\partial_\theta f = \nu e^{-2s}(\partial_\theta^2 + \partial_s^2)f, \qquad (\theta, s) \in \mathbb{T}\times\mathbb{R}.$$

If we localize near a radius $r_0 = e^{s_0}$ where $V'(r_0) \neq 0$, then to leading order, and up to constants, we recover the equation for the Couette flow. In this sense, the Couette flow can be seen as a first local approximation of a radial vortex in $\mathbb{R}^2$, in a region where the angular velocity is strictly monotone. Below is a video showing the effect of the conformal mapping.



Video. Left: a scalar transported and diffused by the circular flow with $V(r) = 1/(1+r^2)$. Right: the same solution in log-polar coordinates, where the flow is a shear flow.
The video plays in the online edition of this page. How it was made: page Numerical methods, record `num-log-polar`.

5.1 The result

Before introducing the main theorem, it is useful to observe that, taking the x-average, one is left with the standard heat equation on $\mathbb{R}$. Namely,

$$\langle f\rangle_x = \frac{1}{2\pi}\int_{\mathbb{T}} f(x,y)\,dx \quad \Longrightarrow \quad \langle f(t)\rangle_x = e^{\nu t\partial_{yy}}\langle f^{\mathrm{in}}\rangle_x.$$

For this part of the solution there is therefore nothing new to say, and it enjoys the standard diffusive estimates. Hence, the quantity to measure is how fast the solution approaches its x-average.

Theorem 1. *Let $\nu \in [0,1)$ and let f be the solution with initial datum $f^{\mathrm{in}} \in L^2(\mathbb{T}\times\mathbb{R})$. There exist constants $c, C > 0$, independent of ν, such that for all $t \geq 0$ the following inequality holds true:*

$$\|(f - \langle f\rangle_x)(t)\|_{L^2} \leq Ce^{-c\nu^{1/3}t}\|f^{\mathrm{in}} - \langle f^{\mathrm{in}}\rangle_x\|_{L^2}.$$

Moreover, if $f^{\mathrm{in}} \in H^1(\mathbb{T}\times\mathbb{R})$, the following mixing estimate, uniform in ν, holds true:

$$\|(f - \langle f\rangle_x)(t)\|_{H^{-1}} \leq \frac{C}{1+t}e^{-c\nu^{1/3}t}\|f^{\mathrm{in}} - \langle f^{\mathrm{in}}\rangle_x\|_{H^1}.$$

The x-average as a projection. The x-average is in fact nothing more than the orthogonal projection in L^2 onto the kernel of the transport operator $y\partial_x$: the functions in the kernel are exactly those that do not depend on x. When the transport operator has a non-trivial kernel, we cannot expect the elements of the kernel to experience any mixing at all from the transport. The interaction with the dissipation is however much more subtle, and it relies heavily on the commutation properties of this projection with the diffusion operator. In this very simple case everything trivializes: the projection commutes with $\nu\Delta$, so the x-average solves the heat equation and $f - \langle f\rangle_x$ solves the same equation as f. But the mathematical structure behind the convergence towards the x-average lies in the fact that the transport operator has a non-trivial kernel.

Why negative Sobolev norms measure mixing. The L^2 norm does not see mixing: for $\nu = 0$ it is conserved, whatever the flow does. The norm of H^{-1} does, because it measures how much of a function has been sent to high frequencies: in Fourier, it weighs each mode by the inverse of the size of its frequency, so it is small when the function lives at high frequencies, even if its L^2 norm is not. On the physical side, negative Sobolev norms scale as powers of a length: the ratio $\|g\|_{H^{-1}}/\|g\|_{L^2}$ can be thought of as the length at which g oscillates. For $\nu = 0$, the theorem says that this length decays like $1/t$. See also the note on mixing in the Overview.

Other function spaces. The theorem is stated in spaces based on L^2. Since the flow is incompressible, the equation does not increase any L^p norm. For $1 < p < \infty$, enhanced dissipation in L^p then follows from the L^2 estimate by interpolation, with a rate that degenerates as $p \to 1$ and as $p \to \infty$. Some techniques give the estimate in L^p directly, without this loss. For the Couette flow, the explicit solution gives the first estimate of the theorem with L^p in place of L^2 for every $p \in [1,\infty]$, with constants independent of p.

Idea. The key idea underlying all the methods below is that the transport operator $y\partial_x$ does not commute with the diffusion $\nu\Delta$: indeed, $[\Delta, y\partial_x] = 2\partial_x\partial_y$. If the two commuted, the solution would be $e^{-ty\partial_x}e^{\nu t\Delta}f^{\mathrm{in}}$ and, since the transport preserves the L^2 norm, the flow would have no effect on the decay. We will highlight in each method where this lack of commutation enters.

At first sight, operators that do not commute look like an annoying technical problem. This is a typical instance where a seemingly unfavorable property, looked at in the right way, turns out to be exactly what helps.

5.2 Notation and setting

Throughout, $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$. For the Fourier transform in x we write

$$f(t,x,y) = \sum_{k\in\mathbb{Z}} f_k(t,y)e^{ikx}, \qquad f_k(t,y) = \frac{1}{2\pi}\int_{\mathbb{T}} f(t,x,y)e^{-ikx}\,dx,$$

so that $f_0 = \langle f\rangle_x$ and $\|f\|_{L^2}^2 = 2\pi\sum_k \|f_k\|_{L^2(\mathbb{R})}^2$. Note that in this model each k-mode evolves independently of the others:

$$\partial_t f_k + iky f_k = \nu(\partial_{yy} - k^2)f_k, \qquad k\in\mathbb{Z}.$$

The Sobolev norms in the theorem are $\|g\|_{H^1}^2 = \|g\|_{L^2}^2 + \|\nabla g\|_{L^2}^2$ and its dual,

$$\|g\|_{H^{-1}} = \sup\Big\{ \int_{\mathbb{T}\times\mathbb{R}} g\varphi\,dx\,dy \;:\; \|\varphi\|_{H^1}\le 1\Big\}.$$

When there is no ambiguity, $\|\cdot\|$ and $\langle\cdot,\cdot\rangle$ denote the norm and the scalar product of $L^2(\mathbb{T}\times\mathbb{R})$ or, for a single mode, of $L^2(\mathbb{R})$.

5.3 One example, every technique

Each proof below is written for this example only, to show the mechanism; the general results, with complete proofs, are in the part *Techniques and general results*.

The estimates mode by mode. Since each k-mode evolves independently, the estimates of the theorem can be stated for each mode separately, keeping track of the dependence on k. There exist constants $c, C > 0$, independent of ν and k, such that for all $k \neq 0$ and $t \ge 0$

$$\|f_k(t)\|_{L^2(\mathbb{R})} \le Ce^{-c\nu^{1/3}|k|^{2/3}t}\|f_k^{\mathrm{in}}\|_{L^2(\mathbb{R})},$$

$$\|f_k(t)\|_{H^{-1}(\mathbb{R})} \le \frac{C}{1+|k|t}e^{-c\nu^{1/3}|k|^{2/3}t}\|f_k^{\mathrm{in}}\|_{H^1(\mathbb{R})},$$

where $H^{\pm1}(\mathbb{R})$ are the Sobolev norms in y only. Higher modes are therefore mixed and dissipated faster. The theorem follows by summing over $k \neq 0$, since $|k| \ge 1$.

5.3.1 Explicit solution

In this proof we take full advantage of the simplicity of the mathematical setting, which is also what allowed Kelvin and Kolmogorov to write down the solution explicitly.

Proof by the explicit solution. For the equation at mode k, we can also take the Fourier transform in y, with the notation

$$\hat f_k(t,\eta) = \int_{\mathbb{R}} f_k(t,y)e^{-i\eta y}\,dy, \qquad f_k(t,y) = \frac{1}{2\pi}\int_{\mathbb{R}} \hat f_k(t,\eta)e^{i\eta y}\,d\eta,$$

so that $\|f\|_{L^2}^2 = \sum_{k\in\mathbb{Z}}\|\hat f_k\|_{L^2(\mathbb{R})}^2$. By elementary properties of the Fourier transform (multiplication by y becomes $i\partial_\eta$), the equation becomes

$$\partial_t\hat f_k - k\partial_\eta\hat f_k = -\nu(k^2+\eta^2)\hat f_k.$$

This is an advection equation in Fourier, with a damping term. We can then follow the characteristics in η, which are the lines $\eta + kt = \text{const}$, and deduce that the explicit solution is given by

$$\hat{f}_k(t,\eta) = \exp\Big(-\nu \int_0^t \big(k^2 + (\eta + k\tau)^2\big)\, d\tau \Big)\, \hat{f}_k^{\text{in}}(\eta + kt).$$

Where the non-commutation enters. In Fourier, the transport is the translation $\eta \mapsto \eta + kt$ and the diffusion is the multiplication by $-\nu(k^2+\eta^2)$. The lack of commutation is simply reflected in the fact that the damping is not constant along the characteristics:

$$\frac{d}{d\tau}\Big[-\big(k^2 + (\eta + k\tau)^2\big)\Big] = -2k(\eta + k\tau),$$

which is exactly the symbol of the commutator $[\Delta, y\partial_x] = 2\partial_x\partial_y$ at the frequency $(k, \eta + k\tau)$. If the two operators commuted, the damping would be constant along the characteristics, the integral in the exponent would be trivial, and we would only recover the decay of the heat equation, as in the idea above. Instead, the frequency in y grows linearly in time along the characteristics, so the damping grows like $\nu k^2\tau^2$ and its integral like $\nu k^2 t^3$.

The only thing left is therefore to estimate the integral in the exponent, for which we claim that

$$\int_0^t \big(k^2 + (\eta + k\tau)^2\big)\, d\tau \geq k^2 t + \frac{1}{12} k^2 t^3 \qquad \text{for all } \eta \in \mathbb{R}.$$

Proof of the claim. Computing the integral and completing the square in η,

$$\int_0^t (\eta + k\tau)^2\, d\tau = \eta^2 t + \eta k t^2 + \frac{1}{3} k^2 t^3 = t\Big(\eta + \frac{kt}{2}\Big)^2 + \frac{1}{12} k^2 t^3 \geq \frac{1}{12} k^2 t^3,$$

with equality only for $\eta = -kt/2$. □

Therefore, using that $f - \langle f \rangle_x = \sum_{k \in \mathbb{Z}\setminus\{0\}} f_k e^{ikx}$, by Plancherel, the fact that $|k| \geq 1$ and that $\nu t^3 \geq 3\nu^{1/3} t - 2$, we get

$$\begin{aligned} \|(f - \langle f \rangle_x)(t)\|_{L^2}^2 &= \sum_{k \neq 0} \int_{\mathbb{R}} e^{-2\nu \int_0^t (k^2 + (\eta + k\tau)^2)\, d\tau}\, |\hat{f}_k^{\text{in}}(\eta + kt)|^2\, d\eta \\ &\leq C^2 e^{-2c\nu^{1/3} t}\, \|f^{\text{in}} - \langle f^{\text{in}} \rangle_x\|_{L^2}^2. \end{aligned}$$

This proves the desired enhanced dissipation estimate. The constants can be explicitly computed.

The constants. Indeed, the claim and $|k| \geq 1$ give the more precise bound

$$\|(f - \langle f \rangle_x)(t)\|_{L^2}^2 \leq e^{-2\nu t - \frac{1}{6}\nu t^3}\, \|f^{\text{in}} - \langle f^{\text{in}} \rangle_x\|_{L^2}^2.$$

Moreover, $s^3 - 3s + 2 = (s-1)^2(s+2) \geq 0$ for $s \geq 0$, which for $s = \nu^{1/3} t$ is the inequality $\nu t^3 \geq 3\nu^{1/3} t - 2$. Hence $e^{-\frac{1}{6}\nu t^3} \leq e^{\frac{1}{3}} e^{-\frac{1}{2}\nu^{1/3} t}$, and one can take $c = 1/4$ and $C = e^{1/6}$.

For the uniform mixing estimate, one can directly use the characterization of the Sobolev norms in Fourier, namely

$$\|g\|_{H^s}^2 = \sum_{k \in \mathbb{Z}} \int_{\mathbb{R}} (1 + k^2 + \eta^2)^s\, |\hat{g}_k(\eta)|^2\, d\eta,$$

which for $s = \pm 1$ are the norms of the theorem. Since $|k| \geq 1$ and $|k|t \leq |\eta| + |\eta + kt|$, by Cauchy–Schwarz we have

$$\begin{aligned}(1+t)^2 \leq (1+|k|t)^2 &\leq \big(1+|\eta|+|\eta+kt|\big)^2 \leq \big(2+\eta^2\big)\big(2+(\eta+kt)^2\big) \\ &\leq \big(1+k^2+\eta^2\big)\big(1+k^2+(\eta+kt)^2\big).\end{aligned}$$

Therefore, bounding the exponential as before,

$$\begin{aligned}\|(f-\langle f\rangle_x)(t)\|_{H^{-1}}^2 &= \sum_{k\neq 0}\int_{\mathbb{R}} \frac{e^{-2\nu\int_0^t (k^2+(\eta+k\tau)^2)\,d\tau}}{1+k^2+\eta^2}\,|\hat f_k^{\mathrm{in}}(\eta+kt)|^2\,d\eta \\ &\leq \frac{C^2 e^{-2c\nu^{1/3}t}}{(1+t)^2}\sum_{k\neq 0}\int_{\mathbb{R}} \big(1+k^2+(\eta+kt)^2\big)|\hat f_k^{\mathrm{in}}(\eta+kt)|^2\,d\eta \\ &= \frac{C^2 e^{-2c\nu^{1/3}t}}{(1+t)^2}\,\|f^{\mathrm{in}}-\langle f^{\mathrm{in}}\rangle_x\|_{H^1}^2,\end{aligned}$$

which is the uniform mixing estimate.

In fact, the mode-by-mode estimates can be obtained in the same way, by keeping track of k.

The solution in physical space. For $\nu > 0$, inverting the Fourier transform in y in the explicit solution is a Gaussian integral, which gives

$$f_k(t,y) = e^{-\nu k^2 t - \frac{1}{12}\nu k^2 t^3}\int_{\mathbb{R}} \frac{1}{\sqrt{4\pi\nu t}}\,e^{-\frac{(y-y')^2}{4\nu t}}\,e^{-ikt\frac{y+y'}{2}}\,f_k^{\mathrm{in}}(y')\,dy'.$$

Summing over k, by the Poisson summation formula we get

$$f(t,x,y) = \int_{\mathbb{T}\times\mathbb{R}} G_t(x,y,x',y')f^{\mathrm{in}}(x',y')\,dx'dy',$$

with

$$\begin{aligned}G_t(x,y,x',y') &= \frac{1}{\sqrt{4\pi\nu t}}\,e^{-\frac{(y-y')^2}{4\nu t}}\sum_{n\in\mathbb{Z}}\frac{1}{\sqrt{2\pi\sigma_t^2}}\exp\Big(-\frac{\big(x-x'-t\frac{y+y'}{2}+2\pi n\big)^2}{2\sigma_t^2}\Big), \\ \sigma_t^2 &= 2\nu t + \frac{1}{6}\nu t^3.\end{aligned}$$

Without the diffusion in x, namely with $\sigma_t^2 = \frac{1}{6}\nu t^3$, and on the whole plane, namely keeping only the term $n = 0$, this is the fundamental solution written by Kolmogorov (1934). In the variable x the solution spreads over the length $\sqrt{\nu t^3}$, much larger than the diffusive one $\sqrt{\nu t}$. For $t \gg \nu^{-1/3}$ this length exceeds the size of the torus and the kernel is almost independent of x: this is the enhanced dissipation seen in physical space.

□

5.3.2 Hypocoercivity

This proof builds on elementary techniques, in the spirit of the hypocoercivity theory of Villani (2009): it only uses energy estimates, namely integrations by parts and the Cauchy–Schwarz inequality. The key point is that the transport does not commute with the operator that generates the dissipation, which here is, morally, ∂_y.

Proof by hypocoercivity. We first simplify the setting by ignoring the diffusion in x. In fact, since ∂_{xx} commutes with all the other operators in the equation, we have $f = e^{\nu t \partial_{xx}} g$, where g solves the equation without the diffusion in x, with $g^{\mathrm{in}} = f^{\mathrm{in}}$, and $e^{\nu t \partial_{xx}}$ does not increase the L^2 norm. It is therefore enough, and somewhat more instructive, to prove the estimate for

$$\partial_t g + y \partial_x g = \nu \partial_{yy} g,$$

with $\nu > 0$, since for $\nu = 0$ there is nothing to prove. Without loss of generality, we also assume that g has zero x-average, since the x-average solves a decoupled equation and a zero x-average is preserved in time. Note that on functions with zero x-average we have the Poincaré inequality in x, namely $\|h\| \le \|\partial_x h\|$.

We now look at the equations satisfied by the derivatives of g. The derivative in x commutes with everything, so $\partial_x g$ solves the same equation as g. The derivative in y does not commute with the transport, since $[\partial_y, y\partial_x] = \partial_x$, and we get

$$\begin{aligned} \partial_t(\partial_x g) + y\partial_x(\partial_x g) &= \nu \partial_{yy}(\partial_x g), \\ \partial_t(\partial_y g) + y\partial_x(\partial_y g) &= \nu \partial_{yy}(\partial_y g) - \partial_x g. \end{aligned}$$

This is a coupled system for $\partial_x g$ and $\partial_y g$, and there is no direct dissipation for $\partial_x g$. However, the last term acts as a forcing in the equation for $\partial_y g$, and this is exactly the effect of the non-commutation between the transport and ∂_y.

> *Relation to hyperbolic–parabolic systems.* Systems of this type, in which only part of the unknowns is directly dissipated, are the typical situation, for instance, in compressible fluids or for the Boltzmann equation. A general framework for such hyperbolic–parabolic systems was developed by Kawashima (1983) in his PhD thesis (see also the references therein). In fact, hypocoercivity and many other modern energy methods have analogies with the techniques employed by Kawashima.

Since the transport is antisymmetric, the energy identity for the first equation is

$$\frac{1}{2}\frac{d}{dt}\|\partial_x g\|^2 = -\nu \|\partial_y \partial_x g\|^2.$$

By the Poincaré inequality in x, the right-hand side clearly controls $\|\partial_y g\|^2$, but it does not directly control $\|\partial_x g\|^2$, because there is no Poincaré inequality when $y \in \mathbb{R}$. To get back some dissipation, one of the key ideas behind hypocoercivity is to exploit mixed inner products. Here,

$$\frac{d}{dt}\langle \partial_x g, \partial_y g\rangle = -\|\partial_x g\|^2 + \text{terms coming from the dissipation.}$$

The game is then to tailor an energy functional containing all the quantities needed to conclude:

- $\|\partial_x g\|$, which is what we want to decay;
- $\langle \partial_x g, \partial_y g\rangle$, which gives the dissipation of $\partial_x g$;
- $\|\partial_y g\|$, which is needed to control the mixed product.

For $\alpha, \beta > 0$ to be chosen, we define

$$E(t) = \frac{1}{2}\Big(\|\partial_x g\|^2 + \alpha \|\partial_y g\|^2 + 2\beta \langle \partial_x g, \partial_y g\rangle\Big).$$

The identities for the last two terms are

$$\frac{1}{2}\frac{d}{dt}\|\partial_y g\|^2 = -\nu\|\partial_{yy} g\|^2 - \langle \partial_x g, \partial_y g\rangle, \qquad \frac{d}{dt}\langle \partial_x g, \partial_y g\rangle = -\|\partial_x g\|^2 - 2\nu \langle \partial_x \partial_y g, \partial_{yy} g\rangle,$$

and therefore

$$\frac{d}{dt}E = -\nu\|\partial_x\partial_y g\|^2 - \alpha\nu\|\partial_{yy} g\|^2 - \beta\|\partial_x g\|^2 - \alpha\langle\partial_x g, \partial_y g\rangle - 2\beta\nu\langle\partial_x\partial_y g, \partial_{yy} g\rangle.$$

The first three terms have a good sign, and the third one is the dissipation of $\partial_x g$ that was missing. The last two have no sign, and must be absorbed by the good ones. By the Cauchy–Schwarz and Young inequalities, and by the Poincaré inequality in x for the second bound,

$$2\beta\nu|\langle\partial_x\partial_y g, \partial_{yy} g\rangle| \le \frac{\nu}{2}\|\partial_x\partial_y g\|^2 + 2\beta^2\nu\|\partial_{yy} g\|^2,$$
$$\alpha|\langle\partial_x g, \partial_y g\rangle| \le \frac{\beta}{2}\|\partial_x g\|^2 + \frac{\alpha^2}{2\beta}\|\partial_x\partial_y g\|^2.$$

Therefore, if

$$2\beta^2 \le \alpha \qquad \text{and} \qquad 2\alpha^2 \le \beta\nu,$$

we obtain

$$\frac{d}{dt}E \le -\frac{\beta}{2}\|\partial_x g\|^2 - \frac{\nu}{4}\|\partial_x\partial_y g\|^2.$$

Moreover, the first condition guarantees that E is comparable to the sum of its first two terms:

$$\frac{1}{4}\|\partial_x g\|^2 \le E \le \frac{3}{4}\|\partial_x g\|^2 + \alpha\|\partial_y g\|^2.$$

The two conditions together force $8\beta^3 \le \nu$. Thus, the dissipation that we can obtain for $\partial_x g$ is at most of order $\nu^{1/3}$, and this is where the rate comes from. We then choose

$$\beta = \frac{1}{2}\nu^{1/3}, \qquad \alpha = \frac{1}{2}\nu^{2/3}.$$

Using once more the Poincaré inequality in x,

$$\frac{d}{dt}E \le -\frac{\nu^{1/3}}{4}\Big(\|\partial_x g\|^2 + \nu^{2/3}\|\partial_y g\|^2\Big) \le -\frac{\nu^{1/3}}{3}E,$$

and hence, for all $t \ge s \ge 0$,

$$\|\partial_x g(t)\|^2 \le 4E(t) \le 4e^{-\frac{1}{3}\nu^{1/3}(t-s)}E(s).$$

The choice of the coefficients. The powers of ν can be guessed as follows. The transport turns derivatives in x into derivatives in y. For $\nu = 0$ the solution is $g^{\mathrm{in}}(x - ty, y)$, so that $\partial_y g \approx -t\partial_x g$ for large times. The dissipation is then expected to satisfy $\nu\|\partial_y g\|^2 \approx \nu t^2\|\partial_x g\|^2$. Once integrated up to a time τ, this is comparable to $\|\partial_x g\|^2$ when $\nu\tau^3 \approx 1$, that is $\tau = \nu^{-1/3}$. At that time the three terms of E have sizes 1, $\alpha\tau^2$ and $\beta\tau$ relative to $\|\partial_x g\|^2$, and they are comparable exactly when $\alpha \approx \nu^{2/3}$ and $\beta \approx \nu^{1/3}$. In other words, E measures ∂_y in units of $\nu^{1/3}$.

It remains to remove $\|\partial_y g^{\mathrm{in}}\|$, which appears in $E(0)$ and is not controlled by the norms stated in the theorem. For this we use the smoothing effect of the diffusion. Integrating in time the energy identity for $\partial_x g$, and using the Poincaré inequality in x, we get

$$\nu\int_0^T \|\partial_y g(s)\|^2\,ds \le \frac{1}{2}\|\partial_x g^{\mathrm{in}}\|^2 \qquad \text{for all } T > 0.$$

We then take $T = \nu^{-1/3}$, the time scale of the decay. Hence there is a time $t_0 \in (0, \nu^{-1/3}]$ such that $\nu^{2/3}\|\partial_y g(t_0)\|^2 \le \frac{1}{2}\|\partial_x g^{\mathrm{in}}\|^2$ and therefore, since $\|\partial_x g\|$ does not increase,

$$E(t_0) \le \frac{3}{4}\|\partial_x g(t_0)\|^2 + \frac{1}{2}\nu^{2/3}\|\partial_y g(t_0)\|^2 \le \|\partial_x g^{\mathrm{in}}\|^2.$$

For $t \geq t_0$ we use the decay of E starting from the time t_0, while for $t \leq t_0$ we simply use that $\|\partial_x g\|$ does not increase. Since $\nu^{1/3} t_0 \leq 1$, in both cases

$$\|\partial_x g(t)\| \leq 2e^{1/6} e^{-\frac{1}{6}\nu^{1/3} t} \|\partial_x g^{\mathrm{in}}\| \qquad \text{for all } t \geq 0.$$

Finally, since ∂_x commutes with the equation and is invertible on functions with zero x-average, the same estimate holds with g in place of $\partial_x g$, and hence for f, since $\|f(t)\| \leq \|g(t)\|$ and $g^{\mathrm{in}} = f^{\mathrm{in}}$. This proves the enhanced dissipation estimate of the theorem.

Avoiding the smoothing step. The smoothing step can be avoided by letting the coefficients depend on time, exactly as one proves the gain of regularity for the heat equation through energy estimates with the weight t in front of $\|\partial_y g\|^2$. With

$$q(t) = \frac{\nu^{1/3} t}{1 + \nu^{1/3} t}, \qquad \alpha(t) = a\, \nu^{2/3} q(t), \qquad \beta(t) = b\, \nu^{1/3} q(t)^2,$$

the functional E at the initial time is just $\frac{1}{2}\|\partial_x g^{\mathrm{in}}\|^2$, so nothing is asked of $\partial_y g^{\mathrm{in}}$. For $t \ll \nu^{-1/3}$ the weights behave like νt and νt^2, as for the heat equation, and for $t \gg \nu^{-1/3}$ they are the constant ones used above. The price is two new terms, coming from the time derivatives of the weights, which are absorbed by the good terms we already have if a and b are small enough (we leave this as an exercise). A functional of this kind measures the solution at each time at the right scale, and gives the decay and the gain of regularity at once.

Uniform mixing estimate. It remains to prove the uniform mixing estimate. We go back to f, which has zero x-average, and use a vector field that commutes with the equation, namely

$$Z := \partial_y + t\partial_x.$$

Indeed, $[\partial_t + y\partial_x, Z] = \partial_x - \partial_x = 0$, and Z has constant coefficients in x and y, so it also commutes with $\nu\Delta$. For $\nu = 0$ we have $Zf(t,x,y) = (\partial_y f^{\mathrm{in}})(x - ty, y)$: Z is the derivative in y in the coordinates that move with the flow. Hence Zf solves exactly the same equation as f, with datum $\partial_y f^{\mathrm{in}}$ and zero x-average, and the enhanced dissipation estimate gives

$$\|Zf(t)\| \leq 2e^{1/6} e^{-\frac{1}{6}\nu^{1/3} t} \|\partial_y f^{\mathrm{in}}\|.$$

To bound the H^{-1} norm we argue by duality. Let $\varphi \in H^1(\mathbb{T} \times \mathbb{R})$. Since f has zero x-average, we can replace φ by $\varphi - \langle\varphi\rangle_x = \partial_x \Phi$, where Φ is the primitive in x with zero x-average. By the Poincaré inequality in x, $\|\Phi\| \leq \|\varphi\|$ and $\|\partial_y \Phi\| \leq \|\partial_y \varphi\|$. Integrating by parts and using $t\partial_x = Z - \partial_y$,

$$t \int_{\mathbb{T}\times\mathbb{R}} f\varphi \, dx\, dy = -\int_{\mathbb{T}\times\mathbb{R}} (t\partial_x f)\Phi \, dx\, dy = -\int_{\mathbb{T}\times\mathbb{R}} (Zf)\Phi \, dx\, dy - \int_{\mathbb{T}\times\mathbb{R}} f\, \partial_y \Phi \, dx\, dy.$$

Therefore $t\|f(t)\|_{H^{-1}} \leq \|Zf(t)\| + \|f(t)\|$. Since also $\|f(t)\|_{H^{-1}} \leq \|f(t)\|$, the enhanced dissipation estimates for f and Zf give

$$(1+t)\|f(t)\|_{H^{-1}} \leq 2\|f(t)\| + \|Zf(t)\| \leq 6e^{1/6} e^{-\frac{1}{6}\nu^{1/3} t} \|f^{\mathrm{in}}\|_{H^1},$$

which is the uniform mixing estimate.

The mode-by-mode estimates can be obtained in the same way: on a single mode k the Poincaré inequality in x becomes the identity $\|\partial_x h\| = |k|\|h\|$, and the argument can be repeated with νk^2 in place of ν in the choice of α, β and T, which gives the rate $\nu^{1/3}|k|^{2/3}$. □

General theory: Hypocoercivity.

5.3.3 Resolvent method

This proof builds on resolvent estimates for the operator $-\nu\partial_{yy} + iky$, uniform along the imaginary axis. In fact, estimates on the decay of the associated semigroup are tied to resolvent bounds by Gearhart–Prüss type theorems, in the spirit of Helffer and Sjöstrand (2010) and Wei (2021). The key point is that ∂_y does not commute with the multiplication by y, so a function cannot concentrate near the critical layer, where iky equals the spectral parameter, unless of course its derivative is large, in which case the dissipation kicks in.

Proof by the resolvent method. We fix $k \neq 0$ and write $f_k = e^{-\nu k^2 t} g$. Then the function g solves

$$\partial_t g + H_k g = 0, \qquad H_k := -\nu\partial_{yy} + iky \quad \text{on } L^2(\mathbb{R}).$$

We note that the spectrum of H_k on $L^2(\mathbb{R})$ is in fact empty (it is a complex Airy operator), and we aim instead at estimating the resolvent of H_k. To this end, we define the *pseudospectral abscissa* of H_k as

$$\Psi(H_k) := \inf \left\{ \|(H_k - ik\lambda)g\| \ : \ \lambda \in \mathbb{R},\ g \in D(H_k),\ \|g\| = 1 \right\}.$$

> *What the pseudospectral abscissa measures.* For $\varepsilon > 0$, the ε-pseudospectrum of H_k is the set of $z \in \mathbb{C}$ where the resolvent is large, $\|(H_k - z)^{-1}\| > 1/\varepsilon$. Equivalently, it is the set of z with an approximate eigenfunction, $\|(H_k - z)g\| < \varepsilon\|g\|$. For a self-adjoint operator this is the ε-neighbourhood of the spectrum. For an operator that is not normal, as H_k, it can be much larger, and it is the resolvent, not the spectrum, that controls the semigroup. The pseudospectral abscissa is the largest ε for which the ε-pseudospectrum does not touch the imaginary axis, namely $\Psi(H_k) = \big(\sup_\lambda \|(H_k - ik\lambda)^{-1}\|\big)^{-1}$. For H_k a translation in y only shifts $\operatorname{Im} z$, so the pseudospectra are half-planes. For more on pseudospectra, see Trefethen and Embree (2005).

This is a fundamental quantity, which controls the decay in time of the semigroups generated by accretive operators (H is accretive if $-H$ is dissipative). Indeed, Gearhart–Prüss type bounds link the pseudospectral abscissa to the decay of the semigroup, see Helffer and Sjöstrand (2010), and here we appeal to the result of Wei (2021) (see the Toolkit), which tells us that

$$\|e^{-tH_k}\|_{L^2 \to L^2} \leq e^{\pi/2}\, e^{-\Psi(H_k)t} \qquad \text{for all } t \geq 0.$$

Therefore, the enhanced dissipation estimate follows from the claim below.

Claim. *There exists $c_0 > 0$, independent of ν and k, such that $\Psi(H_k) \geq c_0\, \nu^{1/3}|k|^{2/3}$.*

Once the claim is proved, the enhanced dissipation estimate of the theorem follows by going back from g to f_k and summing over $|k| \geq 1$.

Proof of the claim. Fix $\lambda \in \mathbb{R}$ and $g \in D(H_k)$, and let

$$F_\lambda := (H_k - ik\lambda)g = -\nu\partial_{yy} g + ik(y - \lambda)g.$$

The goal is to bound $\|g\|$ by $\|F_\lambda\|$, uniformly in λ.

Step 1 (energy identity). The first observation is the standard energy identity, which in this spectral picture reads

$$\operatorname{Re}\big(\langle F_\lambda, g\rangle\big) = \nu\|\partial_y g\|^2, \qquad \text{hence} \qquad \nu\|\partial_y g\|^2 \leq \|F_\lambda\|\|g\|.$$

Note that here we have not used the transport operator at all.

Step 2 (using the transport). To use the transport, we test against $(y-\lambda)g$ and take the imaginary part. Integrating by parts once,

$$\operatorname{Im}\left(\langle F_\lambda, (y-\lambda)g\rangle\right) = k\|(y-\lambda)g\|^2 + \nu \operatorname{Im}\left(\langle \partial_y g, g\rangle\right),$$

and therefore

$$|k|\,\|(y-\lambda)g\|^2 \le \|F_\lambda\|\,\|(y-\lambda)g\| + \nu\|\partial_y g\|\|g\|.$$

This controls g away from the critical layer $y=\lambda$, where the transport vanishes. To recover the full norm of g we use a functional inequality, which only requires an integration by parts:

$$\|g\|^2 = \int_{\mathbb{R}} \partial_y(y-\lambda)\,|g|^2\,dy = -2\operatorname{Re}\left(\langle (y-\lambda)g, \partial_y g\rangle\right) \le 2\,\|(y-\lambda)g\|\,\|\partial_y g\|.$$

In words, g cannot concentrate near the critical layer unless its derivative is large.

Step 3 (conclusion). By Step 2 and Young's inequality,

$$\|(y-\lambda)g\|^2 \le \frac{1}{k^2}\|F_\lambda\|^2 + \frac{2\nu}{|k|}\|\partial_y g\|\|g\|.$$

Inserting this in the functional inequality, and using Step 1 to bound $\|\partial_y g\|$,

$$\|g\|^4 \le 4\,\|(y-\lambda)g\|^2\|\partial_y g\|^2 \lesssim \frac{1}{\nu k^2}\|F_\lambda\|^3\|g\| + \frac{1}{\nu^{1/2}|k|}\|F_\lambda\|^{3/2}\|g\|^{5/2}.$$

By Young's inequality once more, $\|g\|^3 \lesssim \frac{1}{\nu k^2}\|F_\lambda\|^3$, that is $\|F_\lambda\| \gtrsim \nu^{1/3}|k|^{2/3}\|g\|$ uniformly in λ.

A more flexible argument: splitting near and away from the critical layer. The functional inequality of Step 2 uses that $y-\lambda$ has derivative one everywhere. For a general shear $v(y)$ this is not available: for instance $\int v'|g|^2$ has no sign if v is not monotone. One then splits the norm of g, for $\delta>0$ to be chosen.
Near the critical layer the transport is perturbative and only the dissipation is available, but the region is thin:

$$\int_{|y-\lambda|<\delta} |g|^2\,dy \le 2\delta\|g\|_{L^\infty}^2 \le 2\delta\,\|g\|\,\|\partial_y g\|.$$

Away from the critical layer the transport can be inverted:

$$\int_{|y-\lambda|\ge\delta} |g|^2\,dy \le \frac{1}{\delta^2}\|(y-\lambda)g\|^2 \le \frac{1}{k^2\delta^2}\|F_\lambda\|^2 + \frac{2\nu}{|k|\delta^2}\|\partial_y g\|\|g\|.$$

Adding the two,

$$\|g\|^2 \le \frac{1}{k^2\delta^2}\|F_\lambda\|^2 + 2\Big(\delta + \frac{\nu}{|k|\delta^2}\Big)\|\partial_y g\|\|g\|.$$

The two terms in the bracket balance for $\delta = (\nu/|k|)^{1/3}$, the width of the critical layer, and one concludes with Step 1 as before. This is the structure of the proofs for general shear flows, where the test function away from the layer is usually a regularisation of $\operatorname{sgn}(v-\lambda)$; see Gallay (2022) and Coti Zelati and Gallay (2023).

□

Uniform mixing estimate. It remains to prove the uniform mixing estimate. Here it is convenient to use something that commutes, namely the vector field

$$Z_k := \partial_y + ikt.$$

This is a commuting vector field for the transport equation (namely $[Z_k, \partial_t + iky] = 0$), but in this particular case it also commutes with the dissipation. Hence $Z_k f_k$ solves exactly the same equation as f_k, with datum $\partial_y f_k^{\mathrm{in}}$, and by the enhanced dissipation estimate

$$\|(\partial_y + ikt) f_k(t)\|_{L^2} \le C e^{-c\nu^{1/3}|k|^{2/3}t} \|\partial_y f_k^{\mathrm{in}}\|_{L^2}.$$

To bound the H^{-1} norm, we take a test function $\varphi \in H^1(\mathbb{R})$ and note that, integrating by parts,

$$ikt \int_{\mathbb{R}} f_k \varphi \, dy = \int_{\mathbb{R}} (Z_k f_k) \varphi \, dy + \int_{\mathbb{R}} f_k \, \partial_y \varphi \, dy.$$

Therefore, since for $|k|t \le 1$ there is nothing to prove, by the dual definition of the H^{-1} norm and the enhanced dissipation estimates for $Z_k f_k$ and f_k we conclude that, for $|k|t \ge 1$,

$$\|f_k(t)\|_{H^{-1}} \le \frac{1}{|k|t} \Big(\|Z_k f_k(t)\|_{L^2} + \|f_k(t)\|_{L^2} \Big) \lesssim \frac{1}{1+|k|t} \, e^{-c\nu^{1/3}|k|^{2/3}t} \, \|f_k^{\mathrm{in}}\|_{H^1}.$$

□

General theory: Spectral methods.

5.3.4 Stochastic methods

Proof by stochastic methods. [The proof.] □

General theory: Stochastic methods.

5.3.5 Hörmander theory

Proof by Hörmander theory. [The proof.] □

General theory: Hörmander theory.

5.4 What this example teaches

[What the example shows, and what the general theory keeps of it.]

6 Poiseuille flow

Contributors: to be named · Status: skeleton
Discuss on Zulip: #examples, topic "Poiseuille flow".

[Lead: what changes compared with Couette flow, and why.]

6.1 Setting

[The flow, the domain, the critical point, what decays.]

6.2 The result

Theorem 2 (enhanced dissipation for Poiseuille flow)**.** *[The statement.]*

Idea. [The key mechanism in a few lines.]

Proof architecture. [The proof as a short chain of steps, and its diagram.]

6.3 What changes compared with Couette flow

For each technique: what carries over from Couette flow, what is lost, and the proof for this example.

6.3.1 Explicit solution

[What is lost.]

6.3.2 Spectral methods

Proof by spectral methods. [The proof.] □

6.3.3 Hypocoercivity

Proof by hypocoercivity. [The proof.] □

6.3.4 Stochastic methods

Proof by stochastic methods. [The proof.] □

6.3.5 Hörmander theory

Proof by Hörmander theory. [The proof.] □

Part III
Techniques and general results

7 Spectral methods

Contributors: to be named · Status: skeleton
Discuss on Zulip: #techniques, topic "Spectral methods".

[Lead: what the method gives, and under which hypotheses.]

7.1 The method

[The key mechanism, in a few lines.]

Worked example: Couette flow. What changes with a critical point: Poiseuille flow.

7.2 General results

[The theorems, each written as a result (statement, idea, proof architecture, complete proof; see How to contribute); lemmas of general use go to the Toolkit *(Standard analysis, Enhanced-dissipation tools).]*

7.3 Limitations and comparison

[What the method does not give, and how it compares with the other techniques.]

8 Hypocoercivity

Contributors: to be named · Status: skeleton
Discuss on Zulip: #techniques, topic "Hypocoercivity".

[Lead: what the method gives, and under which hypotheses.]

8.1 The method

[The key mechanism, in a few lines.]
Worked example: Couette flow. What changes with a critical point: Poiseuille flow.

8.2 General results

[The theorems, each written as a result (statement, idea, proof architecture, complete proof; see How to contribute); lemmas of general use go to the Toolkit *(Standard analysis, Enhanced-dissipation tools).]*

8.3 Limitations and comparison

[What the method does not give, and how it compares with the other techniques.]

9 Stochastic methods

Contributors: to be named · Status: skeleton
Discuss on Zulip: #techniques, topic "Stochastic methods".

[Lead: what the method gives, and under which hypotheses.]

9.1 The method

[The key mechanism, in a few lines.]
Worked example: Couette flow. What changes with a critical point: Poiseuille flow.

9.2 General results

[The theorems, each written as a result (statement, idea, proof architecture, complete proof; see How to contribute); lemmas of general use go to the Toolkit *(Standard analysis, Enhanced-dissipation tools).]*

9.3 Limitations and comparison

[What the method does not give, and how it compares with the other techniques.]

10 Hörmander theory

Contributors: to be named · Status: skeleton
Discuss on Zulip: #techniques, topic "Hörmander theory".

[Lead: what the method gives, and under which hypotheses.]

10.1 The method

[The key mechanism, in a few lines.]
Worked example: Couette flow. What changes with a critical point: Poiseuille flow.

10.2 General results

[The theorems, each written as a result (statement, idea, proof architecture, complete proof; see How to contribute); lemmas of general use go to the Toolkit *(Standard analysis, Enhanced-dissipation tools).]*

10.3 Limitations and comparison

[What the method does not give, and how it compares with the other techniques.]

Part IV
Extensions and recent advances

11 State of the art

Contributors: to be named · Status: skeleton
Discuss on Zulip: #advances, topic "State of the art".

[The strongest results available to date, in one table.]

Flow class	Domain	Dissipation time	Sharp	Source	Label
—	—	—	—	—	—

12 Related results

Contributors: to be named · Status: skeleton
Discuss on Zulip: #advances, topic "Related results".

Results related to enhanced dissipation and not yet part of the established theory. Each entry gives the statement, a label ([published], [preprint, to be verified] or [raw, unchecked]), the date it was added, its source, and where it would go in the established theory.

12.1 Entries

[First entries.]

12.2 How an entry moves

1. An entry keeps its label until its status changes: raw data checked and posted becomes [preprint, to be verified], a refereed result [published].
2. The editorial board moves established results into the part *Techniques and general results*, or into a new page, with a synthesis proof.

13 Open problems

Contributors: to be named · Status: skeleton
Discuss on Zulip: #advances, topic "Open problems".

Open problems, each with what is known, the expected answer, and the main difficulty.

13.1 Problems

[First problems.]

14 Raw data

Contributors: to be named · Status: skeleton
Discuss on Zulip: #advances, topic “Raw data”.

AI-generated proofs, numerical observations and sketches, labelled [raw, unchecked] until a human has checked them. AI-generated text is quoted verbatim, with its source and date; how an entry moves on is explained on Related results.

14.1 Entries

[First entries.]

Part V
Toolkit

15 Notation

Contributors: to be named · Status: skeleton
Discuss on Zulip: #toolkit, topic “Notation”.

The notation and the definitions used throughout the wiki.

15.1 Notation

[The scalar, the velocity field, the domains, the Fourier modes, the norms.]

15.2 Definitions

[Dissipation time, enhanced dissipation, and the related notions used in the wiki.]

16 Standard analysis

Contributors: to be named · Status: skeleton
Discuss on Zulip: #toolkit, topic “Standard analysis”.

Classical facts of analysis used by the proofs. A statement comes to the Toolkit when several proofs use it, or when it holds in more generality than one result; a step used by one proof only is a claim inside that proof. Each statement is given once, with its proof or a reference, and the proofs elsewhere link to it.

16.1 Inequalities

[Poincaré and interpolation inequalities, and the other inequalities the proofs use.]

16.2 Functional analysis

[Semigroups, spectral theory, and the other facts the proofs use.]

17 Enhanced-dissipation tools

Contributors: to be named · Status: skeleton
Discuss on Zulip: #toolkit, topic "Enhanced-dissipation tools".

Tools specific to enhanced dissipation. A statement comes to the Toolkit when several proofs use it, or when it holds in more generality than one result; a step used by one proof only is a claim inside that proof. Each statement is given once, with its proof or a reference, and the proofs elsewhere link to it.

17.1 Tools

[For instance spectral-gap and resolvent lemmas, and the link between mixing and enhanced dissipation.]

17.1.1 From resolvent bounds to semigroup bounds

Definition 3 (m-accretive operator)**.** A closed operator H on a complex Hilbert space X is m-accretive if, for every $\mu \in \mathbb{C}$ with $\operatorname{Re}\mu > 0$, the operator $H+\mu$ is invertible and $\|(H+\mu)^{-1}\| \le (\operatorname{Re}\mu)^{-1}$. Then $D(H)$ is dense, $\operatorname{Re}\langle Hf, f\rangle \ge 0$ for $f \in D(H)$, and $-H$ generates a semigroup of contractions e^{-tH}.

Theorem 4 (Gearhart–Prüss type bound, Wei)**.** *Let H be an m-accretive operator on X, and let*

$$\Psi(H) := \inf\{\|(H - i\lambda)f\| \ : \ f \in D(H),\ \|f\| = 1,\ \lambda \in \mathbb{R}\}.$$

Then $\|e^{-tH}\| \le e^{\pi/2}\, e^{-\Psi(H)t}$ for all $t \ge 0$.

Idea. Cut the solution in time: $\chi(s)e^{-sH}f$ solves the equation with source $\chi'(s)e^{-sH}f$. After a Fourier transform in time the resolvent on the imaginary axis appears, and by Plancherel the definition of Ψ becomes a weighted inequality for the nonincreasing function $g(s) = \|e^{-sH}f\|^2$. With χ exponential in the bulk this forces g to decay like $e^{-2\Psi s}$. Bringing χ to zero at the two ends costs the constant $e^{\pi/2}$.

Proof. Sources. Wei (2021), Section 2, partly motivated by Helffer and Sjöstrand (2010).

Let $\Psi = \Psi(H)$. Since $D(H)$ is dense, it is enough to prove $\|e^{-tH}f\| \le e^{\pi/2 - \Psi t}\|f\|$ for $f \in D(H)$. Fix such an f and let $g(s) = \|e^{-sH}f\|^2$, which is nonincreasing because the semigroup is a contraction. In particular there is nothing to prove if $\Psi t \le \pi/2$, so we assume $\Psi t > \pi/2$.

Step 1 (an inequality for g).

Claim. *For every $l > 0$ and every real $\chi \in C^1([0,l])$ with $\chi(0) = \chi(l) = 0$,*

$$\int_0^l \chi'(s)^2 g(s)\, ds \ge \Psi^2 \int_0^l \chi(s)^2 g(s)\, ds.$$

Proof of the claim. Let $u(s) = \chi(s)e^{-sH}f$ and $w(s) = \chi'(s)e^{-sH}f$ on $[0,l]$, so that $\partial_s u + Hu = w$. For $\lambda \in \mathbb{R}$ let $\hat{u}(\lambda) = \int_0^l u(s)e^{-i\lambda s}\, ds$, and similarly $\hat{w}$. Since H is closed, $\hat{u}(\lambda) \in D(H)$. Integrating by parts in s, the boundary terms vanish because $\chi(0) = \chi(l) = 0$, and

$$(H + i\lambda)\hat{u}(\lambda) = \hat{w}(\lambda).$$

By the definition of Ψ, $\|\hat{w}(\lambda)\| \ge \Psi\|\hat{u}(\lambda)\|$ for every λ. By Plancherel's theorem for functions with values in X,

$$\int_0^l \|w(s)\|^2\, ds = \frac{1}{2\pi}\int_{\mathbb{R}} \|\hat{w}(\lambda)\|^2\, d\lambda \ge \frac{\Psi^2}{2\pi}\int_{\mathbb{R}} \|\hat{u}(\lambda)\|^2\, d\lambda = \Psi^2 \int_0^l \|u(s)\|^2\, ds.$$

□

Step 2 (choice of χ). Let

$$t_1 = \frac{\pi}{4\Psi}, \qquad t_2 = t - \frac{\pi}{4\Psi}, \qquad t_3 = t + \frac{\pi}{4\Psi}, \qquad l = t + \frac{\pi}{2\Psi},$$

so that $t_1 < t_2$, and

$$\chi(s) = \begin{cases} \sin(\Psi s), & 0 \le s \le t_1, \\ \frac{1}{\sqrt{2}}\, e^{\Psi s - \pi/4}, & t_1 \le s \le t_2, \\ e^{\Psi t - \pi/2} \cos(\Psi(s-t)), & t_2 \le s \le l. \end{cases}$$

This function is C^1 and vanishes at 0 and l. Setting $h := \chi'^2 - \Psi^2 \chi^2$, the claim reads $\int_0^l h\, g\, ds \ge 0$, and

$$h(s) = \begin{cases} \Psi^2 \cos(2\Psi s), & 0 \le s \le t_1, \\ 0, & t_1 \le s \le t_2, \\ -\Psi^2 e^{2\Psi t - \pi} \cos(2\Psi(s-t)), & t_2 \le s \le l. \end{cases}$$

Hence $h \ge 0$ on $[0, t_1]$ and on $[t_3, l]$, $h \le 0$ on $[t_2, t_3]$, and

$$\int_0^{t_1} h\, ds = \frac{\Psi}{2}, \qquad \int_{t_2}^{t} h\, ds = -\frac{\Psi}{2} e^{2\Psi t - \pi}, \qquad \int_t^l h\, ds = 0.$$

Step 3 (conclusion). Since g is nonincreasing, $hg \le h\, g(0)$ on $[0, t_1]$, $hg \le h\, g(t)$ on $[t_2, t]$, and $hg \le h\, g(t_3)$ on $[t, l]$. Therefore

$$0 \le \int_0^l h\, g\, ds \le \frac{\Psi}{2}\, g(0) - \frac{\Psi}{2}\, e^{2\Psi t - \pi} g(t),$$

that is $g(t) \le e^{\pi - 2\Psi t} g(0)$, which is the bound. □

18 Numerical methods

Contributors: to be named · Status: skeleton, three records
Discuss on Zulip: #numerics, topic "Numerical methods".

Every computed picture and video of this wiki has a record on this page: what was computed, how, with which parameters and checks, and the code that produced it.

18.1 How a record works

- A record is a section with an anchor `num-...`; the picture or video links to it with the parameter `record` of Media.
- It gives the equation, the parameters, the method, the checks, the code and how to run it, and its status.
- The code is in the folder `numerics/` of the repository of the wiki, one folder per record, with the output of its checks.

18.2 Records

18.2.1 Heat versus shear on the torus

Produces the video on the Overview (`heat-vs-shear.mp4`) and the check figure below

Equation $\partial_t f + \sin y\, \partial_x f = \nu \Delta f$ on $[0, 2\pi)^2$, periodic, and the heat equation $\partial_t f = \nu \Delta f$; same datum $f_0 = \cos x$

Parameters $\nu = 10^{-3}$, times $0 \le t \le 100$; video of 20 s at 24 frames per second, time running uniformly (5 time units per second), last frame held 1.5 s

Method one Fourier mode in x: $f = \mathrm{Re}(g(t,y)e^{ix})$ with $\partial_t g = -i \sin y\, g + \nu(\partial_y^2 - 1)g$; 512 Fourier modes in y; Strang splitting with time step 0.01, transport and diffusion steps solved exactly; heat equation solved exactly

Checks with 1024 modes and time step 0.005 the norms agree to a relative $6 \cdot 10^{-8}$; the energy balance $\|g(t)\|^2 + 2\nu \int_0^t \|(\partial_y g, g)\|^2 ds = \|g(0)\|^2$ holds to $3 \cdot 10^{-7}$; decay rate of the norm on $[50, 100]$: 0.0167, against $\nu^{1/2} \approx 0.0316$

Code numerics/heat-vs-shear/heat_vs_shear.py in the repository, also below (Python 3, NumPy, Matplotlib, ffmpeg); `python3 heat_vs_shear.py -out ../../wiki/media`, about a minute

Status computed by Claude (AI) on 8 November 2026; not yet checked by a human [demo text]

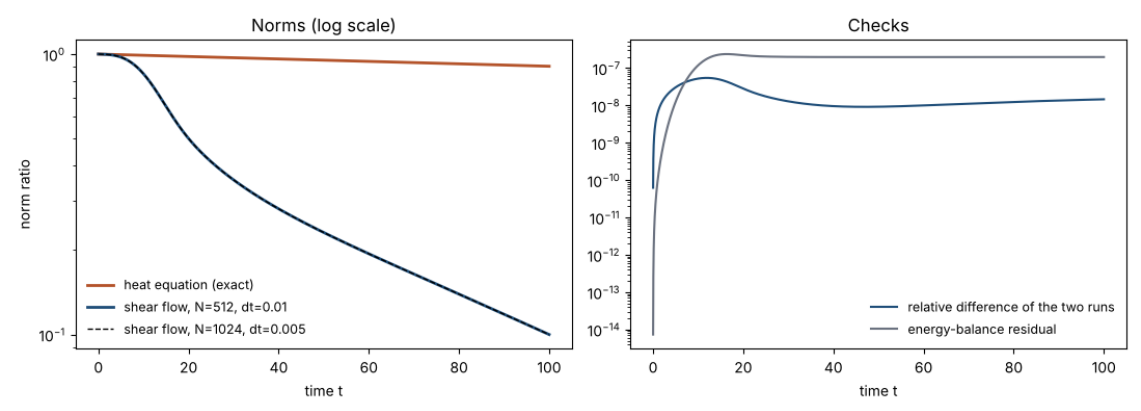


Figure. Left: the norms of the two runs and of the heat equation. Right: the relative difference of the two runs and the energy-balance residual.

Code.

```
#!/usr/bin/env python3
"""Heat equation versus advection-diffusion by a shear flow on the torus [0, 2pi)^2.

    d_t f + sin(y) d_x f = nu (d_xx + d_yy) f,      f(0, x, y) = cos x,

compared with the heat equation (the same without the transport term), whose solution is
exp(-nu t) cos x. The initial datum has zero average in x. It is a single Fourier mode in
    x,
so f(t, x, y) = Re(g(t, y) e^{ix}) with

    d_t g = -i sin(y) g + nu (d_yy - 1) g,      g(0, y) = 1,

solved by a Fourier method in y with Strang splitting: the transport step is the exact
multiplication by exp(-i sin(y) dt), the diffusion step is exact in Fourier variables.
```

```
The L2 norm reported is ||f(t)|| / ||f(0)||, that is the root mean square of |g(t, .)|.

Checks (written to heat-vs-shear-check.txt and heat-vs-shear-check.png):
  * the run is repeated with twice the Fourier modes and half the time step, and the norms
    are compared;
  * energy balance: ||g(t)||^2 + 2 nu int_0^t ||(d_y g, g)||^2 ds = ||g(0)||^2 (means over
    y).

The video lasts --duration seconds; frame s in [0, 1] shows t = T (a s + (1 - a) s^2), a =
    --slow (default 1:
time runs uniformly; a < 1 slows down the start); the last frame is held --hold seconds.

Outputs, in --out: heat-vs-shear.mp4 (video), heat-vs-shear-poster.png (one frame of it),
heat-vs-shear-check.png and heat-vs-shear-check.txt. Needs NumPy, Matplotlib and ffmpeg
(with the libx264 encoder). Run:  python3 heat_vs_shear.py --out ../../wiki/media

Drafted by Claude (AI) on 8 November 2026 for the wiki "Enhanced dissipation for passive
scalars"; record: page Numerical methods, section num-heat-vs-shear.
"""
import argparse
import os
import shutil
import subprocess
import tempfile

import numpy as np
import matplotlib

matplotlib.use("Agg")
import matplotlib.pyplot as plt

HEAT_COLOR = "#b4532a"
SHEAR_COLOR = "#1f4e79"


def solve(nu, T, N, dt, frame_times=()):
    """Times, norms ||f(t)||/||f(0)||, energy-balance residuals, and g(t, y) at
    frame_times."""
    y = 2 * np.pi * np.arange(N) / N
    eta = np.fft.fftfreq(N, d=1.0 / N)              # integer wavenumbers in y
    symbol = eta ** 2 + 1.0                          # symbol of -(d_yy - 1): mode k = 1
    in x
    half = np.exp(-nu * symbol * dt / 2)
    transport = np.exp(-1j * np.sin(y) * dt)
    ghat = np.fft.fft(np.ones(N, dtype=complex))
    steps = int(round(T / dt))
    times = dt * np.arange(steps + 1)
    sq = np.empty(steps + 1)                         # mean over y of |g|^2
    dissip = np.empty(steps + 1)                     # mean over y of |d_y g|^2 + |g|^2
    frames, targets, j = [], list(frame_times), 0
    for n in range(steps + 1):
        power = np.abs(ghat) ** 2 / N ** 2           # Parseval: means over y
        sq[n] = power.sum()
        dissip[n] = (symbol * power).sum()
        while j < len(targets) and targets[j] <= times[n] + dt / 2:
            frames.append(np.fft.ifft(ghat))
            j += 1
        if n == steps:
            break
        ghat *= half
        ghat = np.fft.fft(np.fft.ifft(ghat) * transport) * half
    integral = np.concatenate([[0.0], np.cumsum((dissip[1:] + dissip[:-1]) / 2) * dt])
```

```
    residual = np.abs(sq + 2 * nu * integral - 1.0)
    return times, np.sqrt(sq), residual, frames, y


def field(g, nx):
    """f(x, y) = Re(g(y) e^{ix}) on an nx-by-nx grid (rows: y, columns: x)."""
    step = max(1, len(g) // nx)
    x = 2 * np.pi * np.arange(nx) / nx
    return np.real(g[::step][:, None] * np.exp(1j * x)[None, :])


def draw_frame(path, t, g, nu, times, norms, heat, nx, ymin):
    x = 2 * np.pi * np.arange(nx) / nx
    f_heat = np.exp(-nu * t) * np.cos(x)[None, :] * np.ones((nx, 1))
    f_shear = field(g, nx)
    fig = plt.figure(figsize=(11.2, 7.2), dpi=100)
    gs = fig.add_gridspec(2, 2, height_ratios=[3.6, 1.15], hspace=0.3, wspace=0.06,
                          left=0.08, right=0.97, top=0.845, bottom=0.095)
    for col, (data, title) in enumerate([(f_heat, "Heat equation:  $u = 0$"),
                                         (f_shear, r"Shear flow:  $u = (\sin y,\, 0)$")]):
        ax = fig.add_subplot(gs[0, col])
        ax.imshow(data, origin="lower", extent=[0, 2 * np.pi, 0, 2 * np.pi], cmap="RdBu_r"
    ,
                  vmin=-1, vmax=1, interpolation="bilinear")
        ax.set_title(title, fontsize=17, pad=8)
        ax.set_xticks([]); ax.set_yticks([])
        ax.set_aspect("equal")
    fig.suptitle(r"Same initial datum $f_0 = \cos x$, same diffusivity $\nu = 10^{-3}$"
                 + "        t = %d" % round(t), fontsize=17, y=0.975)
    ax = fig.add_subplot(gs[1, :])
    k = np.searchsorted(times, t, side="right")
    ax.semilogy(times, heat, color=HEAT_COLOR, alpha=0.18, lw=2)
    ax.semilogy(times, norms, color=SHEAR_COLOR, alpha=0.18, lw=2)
    ax.semilogy(times[:k], heat[:k], color=HEAT_COLOR, lw=2.4, label="heat equation")
    ax.semilogy(times[:k], norms[:k], color=SHEAR_COLOR, lw=2.4, label="shear flow")
    ax.axvline(t, color="#888888", lw=1)
    ax.set_xlim(0, times[-1]); ax.set_ylim(ymin, 1.5)
    ax.set_yticks([10.0 ** e for e in range(int(np.log10(ymin)), 1)])
    ax.set_xlabel("time  t", fontsize=13)
    ax.set_ylabel(r"$\|f(t)\|\,/\,\|f_0\|$", fontsize=14)
    ax.tick_params(labelsize=11)
    ax.legend(loc="lower left", fontsize=12, frameon=False, ncol=2)
    fig.savefig(path)
    plt.close(fig)


def main():
    ap = argparse.ArgumentParser(description=__doc__.split("\n")[0])
    ap.add_argument("--out", default=".", help="folder for the video and the figures")
    ap.add_argument("--nu", type=float, default=1e-3)
    ap.add_argument("--T", type=float, default=100.0)
    ap.add_argument("--N", type=int, default=512, help="Fourier modes in y")
    ap.add_argument("--dt", type=float, default=0.01)
    ap.add_argument("--duration", type=float, default=20.0, help="length of the video in
    seconds")
    ap.add_argument("--fps", type=int, default=24)
    ap.add_argument("--slow", type=float, default=1.0, help="speed of time at the start,
    relative to the mean (1: uniform)")
    ap.add_argument("--hold", type=float, default=1.5, help="seconds the last frame is
    held")
    ap.add_argument("--nx", type=int, default=256, help="pixels per side of each panel")
    a = ap.parse_args()
```

```
    os.makedirs(a.out, exist_ok=True)
    s = np.linspace(0, 1, int(round(a.duration * a.fps)))
    frame_times = a.T * (a.slow * s + (1 - a.slow) * s ** 2)
    times, norms, residual, frames, y = solve(a.nu, a.T, a.N, a.dt, frame_times)
    heat = np.exp(-a.nu * times)
    # check: the same run with N doubled and dt halved
    times2, norms2, residual2, _, _ = solve(a.nu, a.T, 2 * a.N, a.dt / 2)
    diff = np.abs(norms2[::2] - norms) / norms2[::2]
    ymin = 10 ** np.floor(np.log10(norms.min()))
    with open(os.path.join(a.out, "heat-vs-shear-check.txt"), "w") as fh:
        fh.write("nu=%g T=%g N=%d dt=%g frames=%d fps=%d\n" % (a.nu, a.T, a.N, a.dt, len(
    frame_times), a.fps))
        fh.write("final norm: shear %.4e, heat %.4e\n" % (norms[-1], heat[-1]))
        fh.write("max relative difference with (2N, dt/2): %.2e\n" % diff.max())
        fh.write("max energy-balance residual: %.2e (2N, dt/2: %.2e)\n" % (residual.max(),
     residual2.max()))
        fh.write("decay rate of the shear norm on [T/2, T]: %.4f; nu^(1/2) = %.4f\n" % (
            np.log(norms[len(norms) // 2] / norms[-1]) / (times[-1] - times[len(times) //
    2]), np.sqrt(a.nu)))
    # check figure
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(11, 4), dpi=100)
    ax1.semilogy(times, heat, color=HEAT_COLOR, lw=2, label="heat equation (exact)")
    ax1.semilogy(times, norms, color=SHEAR_COLOR, lw=2, label="shear flow, N=%d, dt=%g" %
    (a.N, a.dt))
    ax1.semilogy(times2, norms2, color="k", lw=1, ls="--", label="shear flow, N=%d, dt=%g"
     % (2 * a.N, a.dt / 2))
    ax1.set_xlabel("time t"); ax1.set_ylabel("norm ratio"); ax1.legend(frameon=False,
    fontsize=9)
    ax1.set_title("Norms (log scale)")
    ax2.semilogy(times[1:], diff[1:] + 1e-16, color=SHEAR_COLOR, lw=1.5, label="relative
    difference of the two runs")
    ax2.semilogy(times[1:], residual[1:] + 1e-16, color="#6b7280", lw=1.5, label="energy-
    balance residual")
    ax2.set_xlabel("time t"); ax2.legend(frameon=False, fontsize=9); ax2.set_title("Checks
    ")
    fig.tight_layout()
    fig.savefig(os.path.join(a.out, "heat-vs-shear-check.png"))
    plt.close(fig)
    # video
    with tempfile.TemporaryDirectory() as tmp:
        for i, (t, g) in enumerate(zip(frame_times, frames)):
            draw_frame(os.path.join(tmp, "f%04d.png" % i), t, g, a.nu, times, norms, heat,
     a.nx, ymin)
        poster = min(range(len(frame_times)), key=lambda i: abs(frame_times[i] - 0.2 * a.T
    ))
        shutil.copyfile(os.path.join(tmp, "f%04d.png" % poster), os.path.join(a.out, "heat
    -vs-shear-poster.png"))
        subprocess.run(["ffmpeg", "-y", "-loglevel", "error", "-framerate", str(a.fps),
                        "-i", os.path.join(tmp, "f%04d.png"), "-vf", "tpad=stop_mode=clone
    :stop_duration=%g" % a.hold,
                        "-c:v", "libx264", "-pix_fmt", "yuv420p",
                        "-crf", "26", "-preset", "slow", "-movflags", "+faststart",
                        os.path.join(a.out, "heat-vs-shear.mp4")], check=True)
    print(open(os.path.join(a.out, "heat-vs-shear-check.txt")).read().strip())


if __name__ == "__main__":
    main()
```

18.2.2 Alternating shears with random phases

Produces the second video on the Overview (`alternating-shears.mp4`) and the check figure below

Equation $\partial_t f + u \cdot \nabla f = \nu \Delta f$ on $[0, 2\pi)^2$, periodic, with $f_0 = \cos x$ and the alternating shears of Pierrehumbert's model: $u = (\sin(y + \varphi_n), 0)$ for $6n \le t < 6n + 3$ and $u = (0, \sin(x + \psi_n))$ for $6n + 3 \le t < 6n + 6$, the phases independent and uniform on $[0, 2\pi)$; compared with the steady shear $u = (\sin y, 0)$ and with the heat equation. Both velocity fields have size at most 1.

Parameters $\nu = 10^{-4}$, each shear acting for a time 3, times $0 \le t \le 60$, phases from the random generator of NumPy with seed 2026; video of 20 s at 24 frames per second, time running uniformly (3 time units per second), last frame held 1.5 s

Method Fourier pseudo-spectral method on a 512×512 grid; Strang splitting with time step 0.0125, the transport steps (each row or column shifted, through its Fourier coefficients) and the diffusion steps solved exactly; the steady shear as in the record above

Checks with 1024×1024 modes and time step 0.00625 the norms agree to a relative $1.2 \cdot 10^{-6}$; the energy balance $\|f(t)\|^2 + 2\nu \int_0^t \|\nabla f\|^2 ds = \|f_0\|^2$ holds to $3.5 \cdot 10^{-6}$; time for the norm to halve, for $\nu = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}$: alternating shears 8.8, 16.3, 23.9, 31.7 (the same with twice the modes), an increase of about 7.6 for each factor 10 in ν; steady shear 8.6, 20.0, 43.9, 94.9; heat equation $\ln 2/\nu$

Context Exponential mixing for randomly shifted alternating shears is proved by Blumenthal, Coti Zelati and Gvalani (2023), and dissipation on the time scale $|\log \nu|$, with high probability, by Cooperman, Iyer and Son (2025); whether the parameters used here satisfy their hypotheses has not been checked

Code numerics/alternating-shears/alternating_shears.py in the repository (Python 3, NumPy, SciPy, Matplotlib, ffmpeg); `python3 alternating_shears.py -out ../../wiki/media`, about 15 minutes

Status computed by Claude (AI) on 8 November 2026; not yet checked by a human [demo text]

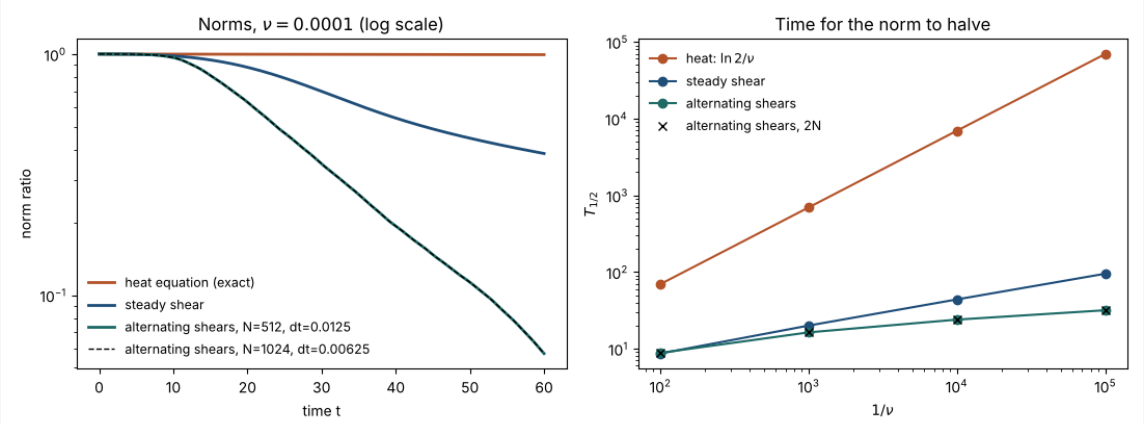


Figure. Left: the norms for $\nu = 10^{-4}$, with two resolutions for the alternating shears. Right: the time for the norm to halve, against $1/\nu$, logarithmic scales.

18.2.3 A vortex in log-polar coordinates

Produces the video in the note *Why this domain?* of Couette flow (`log-polar.mp4`) and the check figure below; the same script draws the figure of that note (`log-polar-map.svg`) from the explicit formulas, without any computation

Equation $\partial_t f + V(r)\partial_\theta f = \nu\Delta f$ in the plane, with $V(r) = 1/(1+r^2)$; in the coordinates $r = e^s$ it is $\partial_t f + V(e^s)\partial_\theta f = \nu e^{-2s}(\partial_\theta^2 + \partial_s^2)f$ on $(\theta, s) \in \mathbb{T}\times\mathbb{R}$; datum $f_0 = \cos(2\theta)\exp(-(s/0.8)^4)$. The checks compare with the same equation without the flow ($V = 0$)

Parameters $\nu = 10^{-3}$, times $0 \le t \le 40$; video of 15 s at 24 frames per second, time running uniformly, last frame held 1.5 s

Method one Fourier mode in θ: $f = \mathrm{Re}(g(t,s)e^{2i\theta})$ with $e^{2s}\partial_t g = -2iV(e^s)e^{2s}g + \nu(\partial_s^2 - 4)g$ for $-3.2 \le s \le 2.8$ and $g = 0$ at the two ends; centred second-order differences with 1500 intervals in s; Crank–Nicolson scheme with time step 0.01

Checks with 3000 intervals and time step 0.005 the norms agree to a relative $2.6\cdot10^{-4}$; the energy balance $\int|g(t)|^2e^{2s}ds + 2\nu\int_0^t\int(|\partial_s g|^2 + 4|g|^2)\,ds\,dt' = \int|g(0)|^2e^{2s}ds$ holds to $1.6\cdot10^{-5}$ of its initial value; $|g|$ at the two ends of the grid stays below $2.2\cdot10^{-5}$. The norm is that of L^2 of the plane, $\|f(t)\|^2 = \int|g(t)|^2e^{2s}ds$ up to a constant; at $t = 40$ it is 0.105 times its initial value, and 0.852 times without the flow

Code numerics/log-polar/log_polar.py in the repository (Python 3, NumPy, SciPy, Matplotlib, ffmpeg); `python3 log_polar.py -out ../../wiki/media`

Status computed by Claude (AI) on 10 November 2026; not yet checked by a human [demo text]

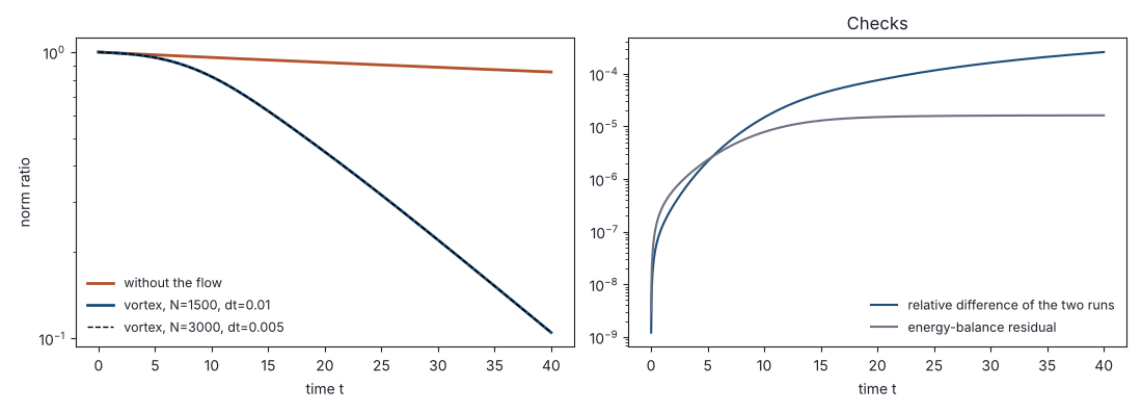


Figure. Left: the norms with the vortex, at two resolutions, and without the flow. Right: the relative difference of the two runs and the energy-balance residual.

18.3 Solvers

[The methods used across the records, with their strengths and limits.]

Part VI
About this wiki

19 How to contribute

Contributors: Michele Dolce · Status: proposal, to be agreed with the editorial board · Text: [demo text]
Discuss on Zulip: #contributing, topic "How to contribute".

This wiki is written from the bottom up, the way a journal works: members of the community propose changes on the talk page of the page they change, proposals are discussed and reviewed there in the open, an associate editor decides, and accepted changes are written into the wiki. The roles, and who holds them, are on Contributors. Short answers to practical questions are in the FAQ at the end of this page. [demo text]

19.1 Roles

- **Editorial board**: the scope and the structure of the wiki; appoints the associate editors; settles discussions that end without consensus.
- **Associate editors**: one or more per part, following the talk pages of its pages and its Zulip channel. They run the discussion of each proposal, ask members of the community for a review when it helps, and decide.
- **Community**: propose, review and discuss. Every member can endorse a new member. Editors can remove a member whose participation disturbs the work rather than helping it.

The first line of each page names its contributors. As the wiki grows, busy parts and pages get more associate editors; the roles stay the same. On the wiki, everyone with a role is a *member* (reads every page and writes on talk pages); the editorial board and the associate editors are also *editors* (edit articles and templates, and write accepted proposals into the pages); administrators give these rights.

19.2 Proposing a change

Every change to a page, from a typo to a new proof, is a *proposal*: a topic on the talk page of the page (tab *Discussion*), which members of the wiki open and discuss. There are three ways to open one.

- *One paragraph, a figure, the central question, or a part still to be written* (a typo, a sentence, a formula, a shorter argument, a caption, a shorter video, a missing proof): in the edition, the pencil button next to it shows its TeX or, for a part still to be written, an empty box (the link *Write this part* opens it too). Change the TeX (theorems, lemmas, remarks, proofs and formulas are written in TeX as usual, with `\begin{lemma} ... \end{lemma}`; the buttons of the box add a note, a folded box as on the Overview, or a claim, a dashed box), write the missing text, or describe the change you ask for (for a picture or a video, for instance), say why, and open the proposal: the wiki shows a new topic on the talk page, already filled in, to be saved. A text too long for a link is copied by the edition, to be pasted into the topic in the wiki.
- *Anything else on a page* (a new box, another order, something to add, to move or to cut): the link *Propose a change* under the title of the page in the edition opens a box. Say in your own words what should change and, if you have it, give the new text in TeX; then

open the proposal: the wiki shows the new topic, already filled in, to be saved. Once it is accepted, an editor writes the change into the page. For a whole new section or a long text, tick the last line of the box and attach the PDF to the topic in the wiki.

- *From the TeX source*, for those who prefer to work on the file: the link *new proposal* under the title of the page in the wiki opens a new topic with a short form. Download the TeX source of the page: link *TeX source* under the title of every page of the edition (with the figures, in a zip, when the page has some). It is a standard LaTeX article that compiles as it is. The TeX of the whole wiki (*TeX of the whole wiki*, on the Overview) is one LaTeX project, for working offline, in which every page is a file of the folder *pages*; such a file can be proposed in the same way. Change what you want and check that it compiles; then paste in the form the blocks you changed (a section, or a statement or a proof with its label), or the whole file: only what changes is applied, and there is no limit of length. Say what the proposal changes, why, and which sources it uses; for a long proposal, attach its PDF (upload it and link it in the form).

Those who are not yet members of the wiki write in the Zulip channel *contributing*, naming the page.

19.2.1 How a proposal is decided

1. The community discusses and reviews the proposal in its topic, with the buttons *Support*, *Oppose* and *Comment* of its box, which add a signed reply (the box counts the supports and the oppositions); replies can also be written by hand, signed with `~~~~`. The author may revise the proposed text in place; the history of the talk page keeps every version.

2. When the discussion has settled, an associate editor of the part decides, with the buttons *Accept* or *Decline* in the box at the top of the topic (shown to editors): the box changes status, and the reason they give is added as a signed reply. There is no deadline. If the discussion ends without consensus, the editorial board decides.

3. An editor writes the accepted proposal into the page with a script. The script converts the TeX to the wiki's format, refuses a paragraph that has changed since the proposal was made, shows exactly what changes on the page, and flags new citations and links. The edit summary links the topic and credits its author, and the box of the topic becomes *applied*, with a link to the change. Decided topics, applied or declined, are moved to the archive of the talk page (its subpage */Archive*), so that the talk page shows only the proposals under discussion or waiting to be applied.

Small corrections (a typo, a broken link, a missing reference) go the same way, and are usually accepted without discussion. Open and accepted proposals are listed in :Category:Open proposals and :Category:Accepted proposals. A proposal changes only the blocks it contains. A paper that is not yet on Literature comes with the proposal that cites it (see *Citations* in The TeX format), and is added to the reference catalogue when the proposal is applied.

19.3 Credits

Under the title of a section, a line in small type names the people who contributed to it, the most recent first: `{{Contributed by|Name Surname, Name Surname}}` on the line after the heading (in TeX, `\contributors{Name Surname, Name Surname}` after the section title). A relevant contribution cannot be measured objectively: the editors decide, at their discretion, who is listed, and add a new name at the beginning of the list. A long list shows the three most recent names, and a link opens the others. The history of each page keeps the record of every change and of its author.

19.4 The TeX format

The TeX source of a page is a standard LaTeX article: one file that compiles with pdflatex and nothing else (on Overleaf, or with TeXShop, TeXstudio, ...; typeset twice for the citations). Its preamble is the same for every page and is never changed; a contribution defines no macros, since the wiki has none. Sections are numbered in the PDF only; the website numbers the results within each page (Theorem 1, Lemma 2, ...).

- Statements: theorem, proposition, lemma, corollary, conjecture, definition, example, remark, claim. The optional argument is the name, the label is the anchor: `\begin{theorem}[name]\label{thm-name}`. Labels start with thm-, prp-, lem-, cor-, def-, clm-, rem- (remarks), cnj- or exm-; a statement sent without a label gets one made from its name or its first words.
- After a statement: the environments idea and architecture, and proofs with an optional title and a label, as in `\begin{proof}[Proof by hypocoercivity]\label{proof-name}`. A claim and its proof sit inside the proof that uses them.
- Formulas: `$...$`, `\[ ... \]` and align*; no numbered equations and no labels on equations; none of the commands that the wiki's renderer rejects (the script lists them).
- Citations: `\citet{Key}` gives *Author (year)* and `\citep{Key}` gives *(Author, year)*, with the keys of Literature; several papers at once: `\citep{Key1,Key2,Key3}`. The reference list at the end of each page is generated from the reference catalogue and contains only the papers that the page cites: a new citation is added to it when the proposal is applied. A paper that is not yet on Literature: cite it with a key of your choice, such as AuthorYear, and give its reference in the field *New references* of the proposal, one line per paper, `\bibitem{AuthorYear} Authors, title, journal volume (year), pages`, with its DOI or arXiv number; it is added to the catalogue and then to the page.
- Links: on the same page, `\hyperref[thm-name]{the theorem above}` rather than `\ref`, since the wiki has no numbers; to another page, `\wikilink{Page\#anchor}{text}`.
- Labels: `\heuristic{}`, `\numerics{}`, `\published{}`, `\preprint{}`, `\raw{}`.
- A new block goes after an existing one: put the line `%% after: Setting` (a section title or a label) just before it.

19.5 Anatomy of a result

A result is a statement followed by its levels of detail, each folded until the reader opens it:

Statement	in the simplest form, constants hidden
Idea	the key mechanism in a few lines
Architecture	the proof as a short chain of steps, later with a diagram
Proof	the complete proof: first its sources (the papers it follows, cited by their key), then the steps; a statement used only in this proof is a *claim*, with its own folded proof inside

A statement that several proofs use, or that holds in more generality than one result, goes to the *Toolkit* (Standard analysis or Enhanced-dissipation tools) and is linked from the proofs. A short proof is written in one piece. The specimen below shows the boxes at work; open the folds to see the nesting.

19.5.1 Specimen

Theorem 5 (specimen)**.** *The statement, in its simplest form.*

Idea. The key mechanism in a few lines, and the heuristic scaling, labelled [heuristic].

Proof architecture. 1. Step 1 reduces the theorem to a claim.

2. Step 2 proves the claim.

3. Step 3 concludes with a lemma of the toolkit.

Proof. Sources. This proof follows ...
Step 1. The reduction.

Claim (specimen)**.** *A statement used only in this proof.*

Proof of the claim. The proof of the claim, folded inside the proof that uses it. □

Step 2. The conclusion, with a lemma of the toolkit. □

19.6 Labels

- [demo text]: text drafted by Claude (AI), until a contributor of the page replaces or approves it.
- [heuristic]: a non-rigorous explanation, kept apart from what is proved.
- [numerics]: a computation, with its record on Numerical methods.
- [published], [preprint, to be verified], [raw, unchecked]: the status of an entry on Related results and Raw data: refereed; posted but not refereed; neither posted nor checked by a human (an AI-generated proof, a numerical observation, a sketch).
- Grey quote boxes: AI-generated text, quoted verbatim with its source and date.

19.7 Provenance and AI use

Where each piece of the wiki comes from, and who has checked what; the labels are those of the section above. [demo text]

- Text drafted by Claude (AI) carries [demo text] until a contributor of the page replaces or approves it.
- Arguments produced by AI systems appear only as verbatim quotations, with source and date, or as entries labelled [raw, unchecked] on Raw data.
- The first line of each page records its contributors, its status and the checks made. A machine check is recorded as such and never counts as a human check.
- Statements attributed to a paper give its theorem number; synthesis proofs say which papers they follow.
- Every accepted proposal is linked, in the summary of the edit that applies it, to the topic of the talk page where it was discussed.

19.7.1 This version of the wiki

On 8 November 2026 the wiki was reorganized into the parts drawn on the map, and its pages were reduced to a skeleton, to be filled by the editorial board and contributors. The first version (6 November 2026: Couette flow with three proofs, a hypocoercivity page, the toolkit, the history) is kept in the project folder, in `archive/2026-11-06-first-version`. The computed pictures and videos, each with its record on Numerical methods, were made by Claude (AI) and are not yet checked by a human. The TeX source of each page is generated from the wiki by a script.

19.7.2 AI-generated documents

Document	Date	Contents	Checks recorded
none	—	—	—

19.7.3 How a result becomes checked

[To be agreed with the editorial board: how many readers a proof needs before it loses its label, and whether a page records who read which proof.]

19.8 Pictures, videos and numerics

A picture, a video or a diagram is one line in the wiki, the template Media; in TeX, `\media{video}{file}{caption}`. The files are kept in the folder `wiki/media` of the project; the edition shows them, the wiki shows a box with the caption. Every computed picture or video has a record on Numerical methods, with the method, the parameters, the checks and the code.

A proposal that adds or replaces a picture or a video names the file in its `\media` line; the file itself is sent separately, since the text of a proposal cannot carry it. A picture (png, svg, jpg or pdf) is uploaded to the wiki and linked in the topic of the proposal. A video (mp4), which the wiki does not accept, is attached to a message in the Zulip topic of the page, where the other members can watch it before the proposal is decided; the topic of the proposal gives the link to that message. For a computed picture or video, attach to the same message the code that produced it, with its parameters and its checks: they make its record on Numerical methods. When the proposal is accepted, an editor adds the files to the edition together with the text. [demo text]

19.9 Discussion and conduct

Proposals are discussed on the talk page of the page they change, so that the discussion stays next to the text and in its history. Zulip is the forum for the mathematics: questions and clarifications, ideas for new results and how they fit the established theory, and larger changes to the structure of the wiki, discussed there before they become proposals. It has one channel per part and one topic per question or idea; the link under the title of each page opens the topic named after the page, for questions about it.

Part	Zulip channel
Overview, Motivation and history	overview
Techniques through examples	examples
Techniques and general results	techniques
Extensions and recent advances	advances
Toolkit	toolkit
Numerical methods	numerics
About this wiki	contributing

Discuss the mathematics, not the people. Associate editors moderate the channel and the talk pages of their part, and can hide messages; editors can remove a member whose participation disturbs the work rather than helping it. The Zulip is endipa.zulipchat.com, by invitation; its administrators invite new members and create channels.

19.10 Formulas in the wiki

For editors who write the wiki's own format.

- Inline formulas need `display="inline"`, displayed ones `display="block"`; each formula is written on one line. The renderer rejects `\newcommand`, `\label`, `\eqref` and `\substack`. No formulas in headings.
- Citations link to the Literature page by the key of the reference catalogue: `[[Literature#BedrossianCotiZelati2017|Bedrossian and Coti Zelati (2017)]]`. The section *References* of each page is generated from the catalogue: add the key to its comment line rather than editing the list.
- To link to a result, use the `id` of its box as the anchor, as in `Couette flow#thm-couette`.

19.11 The edition

A static edition of this wiki, with the pictures, the videos and the TeX source of every page, is built from the wiki's text by a script and never edited by hand: endipa.github.io.

19.12 FAQ

Practical questions, answered briefly; the list grows with the wiki. [demo text]

How do I propose a change to one paragraph?

1. Open the page in the edition and move the pointer over the paragraph, or the figure: a pencil button appears on its right. On a phone or a tablet the button is always shown.
2. Click it: a box opens with the TeX of the paragraph. Change the TeX, and check the result in the preview below it.
3. In the field below the preview, say in one line why. To ask for a change without writing it (a clearer picture, a shorter video), leave the TeX as it is and describe the change in that field. If the text cites a paper that is not on the Literature page, the box names it: give its reference in the field *New references*.
4. Click *Open the proposal on the talk page*. The wiki opens in a new tab (log in if it asks), with a new topic on the talk page of the page, already filled in: check it and click *Save changes*.
5. The other members reply in the topic with *Support*, *Oppose* or *Comment*, and an associate editor accepts or declines. An editor then writes the accepted change into the page, and it appears in the edition when this is next rebuilt.

On a part still to be written, the same button opens an empty box for the missing text. For a whole section, a statement or a proof, see Proposing a change.

20 Contributors

Contributors: editorial board · Status: to be filled
Discuss on Zulip: #contributing, topic “Contributors”.

The people who write and check this wiki, by role; the contributors of each section are also named under its title. How they work together is described in How to contribute.

20.1 Editorial board

Decides the scope and the structure of the wiki, appoints the associate editors, and settles a discussion that ends without consensus.
[Names and roles.]

20.2 Associate editors

Each part of the wiki has at least one associate editor, who follows the talk pages of its pages and its Zulip channel, runs the discussion of every proposal, asks members of the community for a review when it helps, and decides; accepted changes are written into the wiki by an editor. A busy part or page can have several.
[Names, with their parts.]

20.3 Community

Members of the wiki and of the Zulip, who propose, review and discuss changes. Every member can endorse a new member; editors can remove a member whose participation disturbs the work rather than helping it. Early-stage researchers (PhD students, and researchers up to two years after the PhD) can choose to be listed as such.

- Coti Zelati, Michele (Imperial College London)
- Dolce, Michele (EPFL)
- Villringer, David (EPFL)

21 Literature

Contributors: to be named · Status: generated from the reference catalogue (references/catalog.json, checked 2026-11-05)
Discuss on Zulip: #contributing, topic “Literature”.

The papers of the reference catalogue, one line each, grouped by theme; in each theme the newest come first, by the date of their arXiv submission or, without one, of their publication. Each paper is listed once, under its main theme; a page cites what it needs, and its own list of references contains only those papers. This page is generated by a script from a catalogue. A new paper usually arrives with the proposal that cites it, and an entry is corrected by a proposal on the talk page of this page (see How to contribute); the catalogue is updated first. Pages cite these entries by their key and the associated reference appear at the end of the page. The list is a working bibliography, not a survey of the field.

21.1 Reviews

Reviews (3).

- A. L. Mazzucato, Y. Feng and C. Nobili, *Enhanced dissipation by advection and applications to PDEs*, Applied Mathematical Modelling **150** (2026), Part A, 116310,

arXiv:2501.17695.

- M. Coti Zelati, G. Crippa, G. Iyer and A. L. Mazzucato, *Mixing in incompressible flows: transport, dissipation, and their interplay*, Notices of the American Mathematical Society **71** (2024), 593–604, arXiv:2308.00358.
- T. Gallay, *Estimations pseudo-spectrales et stabilité des tourbillons plans (d'après Te Li, Dongyi Wei et Zhifei Zhang)*, Séminaire Bourbaki, exposé n°1167, Astérisque (2022), doi:10.24033/ast.1156.

21.2 Deterministic enhanced dissipation

Shear, radial and other symmetric flows (19).

- S. He, *Localized Enhanced Dissipation: A Hypocoercivity Approach*, preprint (2026), arXiv:2603.14657.
- J. Benthaus, G. M. Coclite and C. Nobili, *Mixing and enhanced dissipation in a time-translating shear flow*, preprint (2026), arXiv:2603.14624.
- J. Benthaus and C. Nobili, *Enhanced dissipation via time-modulated velocity fields*, Evolution Equations and Control Theory **15** (2026), 21–50, arXiv:2501.16905.
- Y. Feng and A. L. Mazzucato, *Dissipation enhancement by shear flows for generalized diffusion operators in high dimension*, Communications in Mathematical Sciences **23** (2025), 2203–2215, doi:10.4310/CMS.250930031923.
- J. Bedrossian, S. He, S. Iyer and F. Wang, *Pseudo-Gevrey smoothing for the passive scalar equations near Couette*, Journal of Functional Analysis **289** (2025), 110987, arXiv:2405.19233.
- D. Coble and S. He, *A note on enhanced dissipation of time-dependent shear flows*, Communications in Mathematical Sciences **22** (2024), 1685–1700, arXiv:2309.15738.
- M. Coti Zelati, M. Dolce and C. Lo, *Diffusion enhancement and Taylor dispersion for rotationally symmetric flows in discs and pipes*, Journal of Mathematical Fluid Mechanics **26** (2024), Article 12, arXiv:2305.18162.
- Y. Feng, A. L. Mazzucato and C. Nobili, *Enhanced dissipation by circularly symmetric and parallel pipe flows*, Physica D: Nonlinear Phenomena **445** (2023), 133640, doi:10.1016/j.physd.2022.133640.
- M. Coti Zelati and T. Gallay, *Enhanced dissipation and Taylor dispersion in higher-dimensional parallel shear flows*, Journal of the London Mathematical Society **108** (2023), 1358–1392, arXiv:2108.11192.
- D. Albritton, R. Beekie and M. Novack, *Enhanced dissipation and Hörmander's hypoellipticity*, Journal of Functional Analysis **283** (2022), 109522, arXiv:2105.12308.
- S. He, *Enhanced dissipation, hypoellipticity for passive scalar equations with fractional dissipation*, Journal of Functional Analysis **282** (2022), 109319, arXiv:2103.07906.
- M. Coti Zelati and T. D. Drivas, *A stochastic approach to enhanced diffusion*, Annali della Scuola Normale Superiore di Pisa, Classe di Scienze **22** (2021), 811–834, arXiv:1911.09995.

- M. Coti Zelati, *Stable mixing estimates in the infinite Péclet number limit*, Journal of Functional Analysis **279** (2020), 108562, arXiv:1909.01310.
- M. Coti Zelati and M. Dolce, *Separation of time-scales in drift-diffusion equations on R^2*, Journal de Mathématiques Pures et Appliquées **142** (2020), 58–75, arXiv:1907.04012.
- M. Beck, O. Chaudhary and C. E. Wayne, *Rigorous justification of Taylor dispersion via center manifolds and hypocoercivity*, Archive for Rational Mechanics and Analysis **235** (2020), 1105–1149, arXiv:1804.06916.
- J. Bedrossian and M. Coti Zelati, *Enhanced dissipation, hypoellipticity, and anomalous small noise inviscid limits in shear flows*, Archive for Rational Mechanics and Analysis **224** (2017), 1161–1204, arXiv:1510.08098.
- M. Beck, O. Chaudhary and C. E. Wayne, *Analysis of enhanced diffusion in Taylor dispersion via a model problem*, Hamiltonian Partial Differential Equations and Applications, Fields Institute Communications **75** (2015), pp. 31–71, arXiv:1501.05877.
- A. Kolmogorov, *Zufällige Bewegungen (Zur Theorie der Brownschen Bewegung)*, Annals of Mathematics (2) **35** (1934), 116–117, doi:10.2307/1968123.
- W. Thomson (Lord Kelvin), *Stability of fluid motion: rectilineal motion of viscous fluid between two parallel planes*, Philosophical Magazine (5) **24** (1887), 188–196, doi:10.1080/14786448708628078.

Other smooth flows (4).

- M. Dolce, C. J. P. Johansson and M. Sorella, *Dissipation enhancing properties for a class of Hamiltonian flows with closed streamlines*, Communications in Partial Differential Equations **50** (2025), 301–352, arXiv:2407.06884.
- T. M. Elgindi, K. Liss and J. C. Mattingly, *Optimal enhanced dissipation and mixing for a time-periodic, Lipschitz velocity field on T2*, Duke Mathematical Journal **174** (2025), 1209–1260, arXiv:2304.05374.
- E. Bruè, M. Coti Zelati and E. Marconi, *Enhanced dissipation for two-dimensional Hamiltonian flows*, Archive for Rational Mechanics and Analysis **248** (2024), Article 84, arXiv:2211.14057.
- G. Iyer and H. Zhou, *Quantifying the dissipation enhancement of cellular flows*, SIAM Journal on Mathematical Analysis **55** (2023), 6496–6516, arXiv:2209.11645.

Rough velocity fields (5).

- M. Romito and L. Roveri, *Instantaneous Total Enhanced Dissipation For Very Rough Shear Flows*, preprint (2025), arXiv:2510.27331.
- T. M. Elgindi and K. Liss, *Norm growth, non-uniqueness, and anomalous dissipation in passive scalars*, Archive for Rational Mechanics and Analysis **248** (2024), arXiv:2309.08576.
- L. Galeati and M. Gubinelli, *Mixing for generic rough shear flows*, SIAM Journal on

Mathematical Analysis **55** (2023), 7240–7272, arXiv:2107.12115.

- M. Colombo, M. Coti Zelati and K. Widmayer, *Mixing and diffusion for rough shear flows*, Ars Inveniendi Analytica, Paper No. **2** (2021), 21 pp., arXiv:2009.12268.
- T. D. Drivas, T. M. Elgindi, G. Iyer and I. Jeong, *Anomalous dissipation in passive scalar transport*, Archive for Rational Mechanics and Analysis **243** (2022), 1151–1180, arXiv:1911.03271.

General criteria and limits (11).

- W. Cooperman, G. Iyer, K. Rowan and S. Son, *Exponentially mixing flows with slow enhanced dissipation*, preprint (2025), arXiv:2507.21305.
- C. Nobili and S. Pottel, *Lower bounds on mixing norms for the advection diffusion equation in Rd*, Nonlinear Differential Equations and Applications NoDEA **29** (2022), Article 12, arXiv:2006.04614.
- C. Seis, *Bounds on the rate of enhanced dissipation*, Communications in Mathematical Physics **399** (2023), 2071–2081, arXiv:2003.08794.
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- M. Coti Zelati, M. G. Delgadino and T. M. Elgindi, *On the relation between enhanced dissipation timescales and mixing rates*, Communications on Pure and Applied Mathematics **73** (2020), 1205–1244, arXiv:1806.03258.
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- C. Villani, *Hypocoercivity*, Memoirs of the American Mathematical Society **202** (2009), no. 950, arXiv:math/0609050.
- P. Constantin, A. Kiselev, L. Ryzhik and A. Zlatoš, *Diffusion and mixing in fluid flow*, Annals of Mathematics **168** (2008), 643–674, arXiv:math/0509663.
- L. N. Trefethen and M. Embree, *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*, Princeton University Press (2005), doi:10.1515/9780691213101.
- S. Kawashima, *Systems of a Hyperbolic-Parabolic Composite Type, with Applications to the Equations of Magnetohydrodynamics*, Doctoral thesis, Kyoto University (1983), doi:10.14989/doctor.k3193.

21.3 Stochastic enhanced dissipation

Random velocity fields (2).

- W. Cooperman, G. Iyer and S. Son, *A Harris theorem for enhanced dissipation, and an example of Pierrehumbert*, Nonlinearity **38** (2025), 045027, arXiv:2403.19858.

- J. Bedrossian, A. Blumenthal and S. Punshon-Smith, *Almost-sure enhanced dissipation and uniform-in-diffusivity exponential mixing for advection-diffusion by stochastic Navier-Stokes*, Probability Theory and Related Fields **179** (2021), 777–834, arXiv:1911.01561.

Transport noise (2).

- F. Flandoli and D. Luo, *Enhanced dissipation and Lyapunov exponents for stochastic transport-diffusion equations with small molecular diffusivity and small noise intensity*, Stochastics and Partial Differential Equations: Analysis and Computations **14** (2026), 1062–1076, doi:10.1007/s40072-026-00426-y.
- M. Coti Zelati, M. Hairer and D. Villringer, *A Stochastic RAGE Theorem and Enhanced Dissipation for Transport Noise*, preprint (2025), arXiv:2507.11422.

21.4 Deterministic mixing

Deterministic mixing (4).

- G. Crippa, T. Elgindi, G. Iyer and A. L. Mazzucato, *Growth of Sobolev norms and loss of regularity in transport equations*, Philosophical Transactions of the Royal Society A **380** (2022), 20210024, arXiv:2109.14975.
- T. M. Elgindi and A. Zlatoš, *Universal mixers in all dimensions*, Advances in Mathematics **356** (2019), 106807, arXiv:1809.09614.
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21.5 Stochastic mixing

Stochastic mixing (2).

- A. Del Zotto and M. Nualart, *Exponential mixing and enhanced dissipation on the unit sphere with Rossby-Haurwitz flows*, preprint (2026), arXiv:2606.11055.
- A. Blumenthal, M. Coti Zelati and R. S. Gvalani, *Exponential mixing for random dynamical systems and an example of Pierrehumbert*, Annals of Probability **51** (2023), 1559–1601, arXiv:2204.13651.

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M. Coti Zelati and T. Gallay, *Enhanced dissipation and Taylor dispersion in higher-dimensional parallel shear flows*, Journal of the London Mathematical Society **108** (2023), 1358–1392, arXiv:2108.11192.

T. Gallay, *Estimations pseudo-spectrales et stabilité des tourbillons plans (d'après Te Li, Dongyi Wei et Zhifei Zhang)*, Séminaire Bourbaki, exposé n°1167, Astérisque (2022), doi:10.24033/ast.1156.

B. Helffer and J. Sjöstrand, *From resolvent bounds to semigroup bounds*, preprint (2010), arXiv:1001.4171.

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